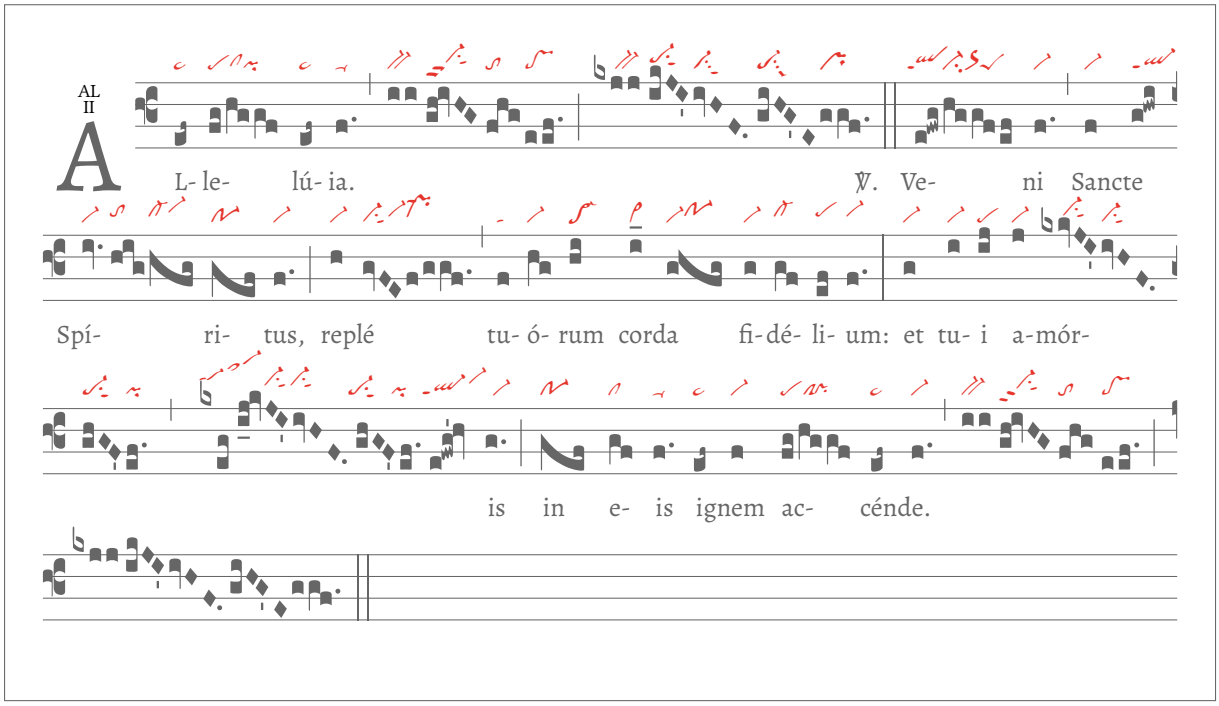
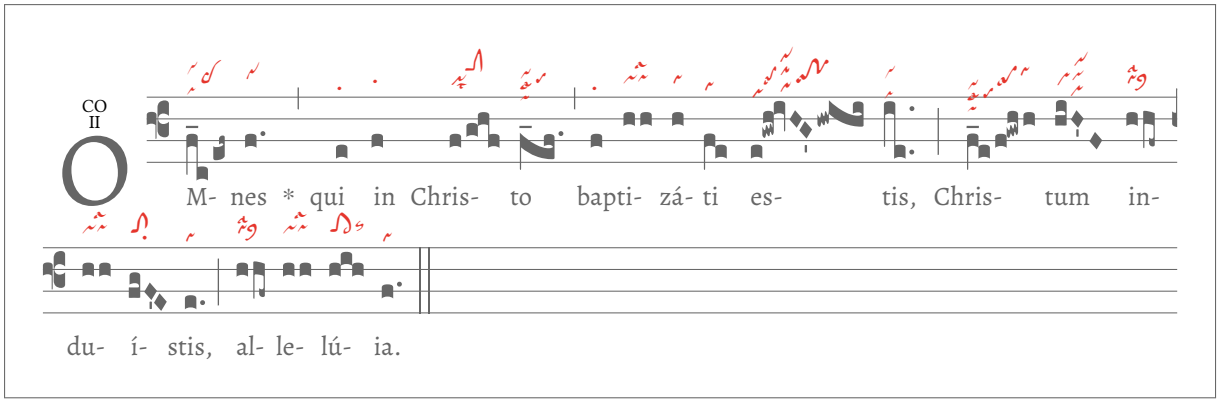
nabc language for gregorio





The nabc language provides the ability to describe some adiastematic neumes, for now just the St. Gallen and Laon (Metz notation family) style. The language is partially based on Dom Eugène Cardine's Table of neumatic signs, but for more complex neumes doesn't always match how the neumes are called; instead attempts to make it easier to compose complex neumes from basic glyphs. To describe adiastematic neumes in gabc, the header should contain `nabc-lines: 1;` line, like:

```
nabc-lines: 1;
(f3) AL(ef~|ta>)le(fg/hggf|peclhgpi)lú(ef~|ta>)ia.(f.|ta-) (,)
(ii//|bv-|gh!ivHG//|vi-hhppu2su1sut1|fhg/|to|eef.|pt) (;)
```

The nabc snippets are then separated by `|` character from gabc snippets or other nabc snippets. Every gabc snippet may be followed by multiple nabc snippets. The maximum number of consecutive nabc snippets is the number declared in the header field `nabc-lines: x;`. After reaching that number of consecutive nabc snippets another gabc snippet followed by nabc snippets can follow. A single nabc snippet is not split into multiple lines, so for larger melismatic pieces it is desirable to synchronize the gabc snippets with corresponding nabc snippets. With `nabc-lines: 1;` the gabc and nabc snippets form an alternating pattern, like `(gabc|nabc|gabc|nabc|gabc)`, in this case the last gabc snippet does not have any corresponding nabc neumes. With `nabc-lines: 2;` the snippets ordering could be e.g. `(gabc|nabc1|nabc2|gabc|nabc1)`.
Each nabc snippet consists of a sequence of *complex neume descriptors*. Each *complex neume descriptor* consists of optional *horizontal spacing adjustment descriptor*, then *complex glyph descriptor*, optionally followed by a sequence of *subpunctis and prepunctis descriptor*s, optionally followed by a sequence of *significant letter descriptor*s.
The *horizontal spacing adjustment descriptor* consists of a sequence of horizontal spacing adjustment characters `/` and `` ` ``.

- `//` move by `nabclargerspace` skip to the right
- `/` move by `nabcinterelementspace` skip to the right
- `` `` `` move by `nabclargerspace` skip to the left
- `` ` `` move by `nabcinterelementspace` skip to the left

The *complex glyph descriptor* consists of a *glyph descriptor*, optionally followed by a sequence of other *glyph descriptor*s, all separated by the `!` character. This is used to describe more complex glyphs, where certain basic glyphs are connected together.
Each *glyph descriptor* consists of a *basic glyph descriptor*, followed by optional *glyph modifiers*, followed by optional *pitch descriptor*.
The following description is for the St. Gall family of neumes, when using the gregall or gresgmodern fonts. For details on Laon (Metz notation family) neumes see 4.
The *basic glyph descriptor* is a two-letter string from the following table:

`vi` virga	`po` porrectus	`tr` torculus resupinus	`tv` trivirga	`pq` pes quassus
`pu` punctum	`to` torculus	`st` stropha	`pr` pressus maior	`ql` quilisma (3 loops)
`ta` tractulus	`ci` climacus	`ds` distropha	`pi` pressus minor	`qi` quilisma (2 loops)
`gr` gravis	`sc` scandicus	`ts` tristropha	`vs` virga strata	`pt` pes stratus
`cl` clivis	`pf` porrectus flexus	`tg` trigonus	`or` oriscus	
`pe` pes	`sf` scandicus flexus	`bv` bivirga	`sa` scandicus	

The `glyph modifiers` is a possibly empty sequence of following characters optionally followed by a number:

- `S` modification of the mark
- `G` modification of the grouping (neumatic break)
- `M` melodic modification
- `-` addition of episema
- `>` augmentive liquescence
- `~` diminutive liquescence

If Dom Cardine's table contains multiple glyphs with the same modifiers, a positive number is added afterwards. E.g. for augmentive liquescent clivis the table shows two different glyphs, the ancus `cl>` and then another neume - `cl>1`, the first neume does not contain any number after it, while the `1` indicates first variant.
The *pitch descriptor* allows to specify the vertical position of the neume. There are no staves, so the vertical position is only rough. For pitches the same letters as in gabc are used, `a` through `n` and `p`. If the *pitch descriptor* is missing, the default is `hf`, otherwise it consists of the letter `h` followed by the pitch letter. Within the *complex glyph descriptor*, each *basic glyph descriptor* has its own pitch, but in the current fonts there are no glyphs with different relative pitches, so if you use a *pitch descriptor* on any of the *basic glyph descriptor*s in the *complex glyph descriptor*, it is best to use the same one on all the other *basic glyph descriptor*s in the same *complex glyph descriptor*.
The *subpunctis and prepunctis descriptor* consists of the letters `su` for subpunctis or `pp` for prepunctis, followed optionally by a modifier letter from the following table and finally a mandatory positive number of repetitions. If the modifier letter is missing, it is a punctum.

- `t` tractulus

- `u` tractulus with episema
- `v` tractulus with double episema
- `w` gravis
- `x` liquescens stropha
- `y` gravis with episema

Only subpunctis are normally used in neume classification, the prepunctis is a `nabc` concept to describe the rising sequence of punctis, tractulis etc. in the left low corner of some neume. While e.g. `vipp2` describes the same neume as `sc`, the former form allows better control on how many punctis or tractulis or tractulis with episema etc. there are. Some examples: `ppt3` stands for 3 raising tractulis with episema, while `su1sut1sux1` stands for a punctum, followed by tractulus, followed by liquescens stropha.
The *significant letter descriptor* consists of the letters `ls`, followed by a shorthand of the significant letter or common group of them, followed by a number - a rough position relative to the *complex glyph descriptor* glyph with prepunctis and subpunctis attached to it. If more than one *significant letter descriptor* is used on the same *complex neume descriptors* for the same position, then they are ordered in the order they are written in the `nabc` snippet at that position. The position numbers are:

`1` left upper corner	`2` above the neume	`3` right upper corner
`4` to the left of the neume		`6` to the right of the neume
`7` left bottom corner	`8` below the neume	`9` right bottom corner

The shorthands, including the `ls` prefix, which should be followed by the above mentioned position digit:

`lsal` altius	`lsiv` iusum valde	`lss` sursum
`lsam` altius mediocriter	`lsk` klenche	`lssb` sursum bene
`lsb` bene	`lsl` levare	`lssc` sursum celeriter
`lsc` celeriter	`lslb` levare bene	`lssimil` similiter
`lscm` celeriter mediocriter	`lslc` levare celeriter	`lssimul` simul
`lsco` coniunguntur	`lslen` leniter	`lssm` sursum mediocriter
`lscw` celeriter (wide form)	`lslm` levare mediocriter	`lsst` sursum tenere
`lsd` deprimatur	`lslp` levare parvum	`lssta` statim
`lse` equaliter	`lslt` levare tenere	`lst` tenere
`lseq` equaliter	`lsm` mediocriter	`lstb` tenere bene
`lsew` equaliter (wide form)	`lsmoll` molliter	`lstm` tenere mediocriter
`lsfid` fideliter	`lsp` parvum	`lstw` tenere (wide form)
`lsfr` frendor	`lspar` paratim	`lsv` valde
`lsg` gutture	`lspfec` perfecte	`lsvol` volubiliter
`lsi` iusum	`lspm` parvum mediocriter	`lsx` expectare
`lsim` iusum mediocriter	`lspulcre` pulcre	

Finally a commented example of a `nabc` snippet:

```
```po////pe>2lse7lsl3qlhh!vshhppt1sut2ql>ppu3
```

``` stands for a negative horizontal skip (move to the left) by `nabcinterelementspace` and `nabclargerspace`.
`po` is a basic porrectus, at the default pitch (height `hf`).
`////` stands for a horizontal skip (move to the right) by two times `nabclargerspace`.
`pe>2lse7lsl3` stands for the 3rd augmentive liquescent form of podatus (epiphonus with tractulus, see the following page) at the default pitch, with equaliter in the low left corner and levare in the upper right corner.
`qlhh!vshhppt1sut2` stands for a 3 loop quilisma joined with virga strata, both at relative pitch 2 above the default one, with a single tractulus before it and two subpunctis - tractulis.
`ql>ppu3` is a 3 loop liquescent quilisma with 3 tractulis with episema before it.

The *basic glyph descriptor* for Laon neumes is very similar to the one used for St. Gall neumes, but with uncinus and oriscus-clivis added and stropha, 2 loops quilisma and gravis removed:

`un` uncinus
`vi` virga
`pu` punctum
`ta` tractulus
`gr` gravis
`cl` clivis
`oc` oriscus-clivis

`pe` pes
`po` porrectus
`to` torculus
`ci` climacus
`sc` scandicus
`pf` porrectus flexus
`sf` scandicus flexus

`tr` torculus resupinus
`ds` distropha
`ts` tristropha
`tg` trigonus
`bv` bivirga
`tv` trivirga
`pr` pressus maior

`pi` pressus minor
`vs` virga strata
`or` oriscus
`sa` scandicus
`pq` pes quassus
`ql` quilisma
`pt` pes stratus

The *subpunctis and prepunctis descriptor* consists of the letters `su` for subpunctis or `pp` for prepunctis, followed optionally by a modifier letter from the following table and finally a mandatory positive number of repetitions. If the modifier letter is missing, it is a punctum.

- `n` uncinus
- `q` quilisma
- `z` virga
- `x` cephalicus

The *significant letter descriptor* for Laon neumes allows another position, `5`, which stands for inside the neume. The exact position of the letter depends on the glyph, if there is no glyph with the neume inside of it, it can't be positioned inside the neume.
The shorthands, including the `ls` prefix, which should be followed by the above mentioned position digit for Laon neumes are:

`lsa` augete
`lsc` celeriter
`lseq` equaliter
`lseq-` equaliter
`lsequ` equaliter
`lsf` fastigium
`lsh` humiliter
`lshn` humiliter nectum
`lshp` humiliter parum
`lsl` levare
`lsn` non (tenere), negare, nectum, naturaliter

`lsnl` non levare
`lsnt` non tenere
`lsm` mediocriter
`lsmd` mediocriter
`lss` sursum
`lssimp` simpliciter
`lssimpl` simpliciter
`lssp` sursum parum
`lsst` sursum tenere
`lst` tenere
`lsth` tenere humiliter

For the Tironian notes the form of the *significant letter descriptor* starts with the `lt` prefix followed by letters from the following list, followed by above mentioned position digit (`5` can't be used):

`lti` iusum
`ltdo` deorsum
`ltdr` devertit
`ltdx` devexum
`ltps` prode sub eam (trade subtus)
`ltqm` quam mox
`ltsb` sub
`ltse` seorsum

`ltsj` subjice
`ltsl` saltim
`ltsn` sonare
`ltsp` supra
`ltsr` sursum
`ltst` saltate (salte)
`ltus` ut supra

Names of Symbols	Simple Symbols	Symbols differentiated by				Symbols indicating a special meaning		
		the addition		the modification		melodic	phonetic	
		of letters	of episemas	of the mark	of the grouping (neumatic break)		liquescence	
							augment.	dimi nut.
virga	vi	vilsc3 vilst1 vippt1lsc2	vi-				vi>	
tractulus	ta	talsc3 talst2					ta>	
punctum	pu						st visu1sux1	
gravis	gr							
clivis	cl	cllsc3 cllst2 cl-lst2	cl- clS-	clS clS1		clM	cl> cl>1	cl~
pes	pe		pe- pe-1	peS	peG	peG	pe> pe>1 pe>2	pe~
porrectus	po	polsc2	po- po-1		poG		po>	po~
torculus	to	tolsc3 tolsc4lst3 tolst2	to- to-1	toS toS1 toS2 toS3	toG	toM toM1	to>	pe>
climacus	ci ci1	cilsc3 cilst2	ci-	ciS ciS1 visut1su2	ciG ciG1	ciM	ci>	ci~
scandicus	sc		sc-	scS	scG scG1	scM1	sc>	sc~
	sc1		vi-hevi-hivi-hm			vihevihivihm		
porrectus flexus	pf	pflsc2 pf1lsc2 pflst2	pf-	pfS	sfM	pfM	pf>	pf~
pes subbipunctis	pesu2	vippt1su2lsc3	pe-su2 pe-1su2	peSsu2 peSsut2	vippt1su2	vi-!ci pesu1suw1	pesu1sux1	to>
			pe-1sut2		toS2sut1			
scandicus flexus	sf	sflsc3	sf-	sfS	vi-tohk peclhi	sfM	cl>ppt2	sc>
torculus resupinus	tr	trlsc2 trlst2	tr-1	trS	trG trG1 trG2	trM	tr>	tr~
apostropha	st		st-	stS			st>	
distropha	ds	ds-lsx6	ds-				ds>	
tristropha	ts	ts-lsx6	ts-			tsM	ts>	
trigonus	tg tg1			tgS tgS1 tgS3		tgM		ds>
bigirga & trivirga	bv tv		bv- tv-				bv>	
pressus	pr pi	prlsc2 pr-lsc2 prlst2	pr-			prM	pr> pi>	vs>
		pilsc2 pilst2						or>
virga strata	vs	vslsc3 vslst2	vs-				vs>	ta>
oriscus	or						or>	
salicus	sa sa1	salsfr1 saM1lsfr1	sa- sa-1		saG	saM saM1 saM2	sa> sa->1	pe>2
pes quassus	pq	pqlsfr1	pq-				pq>	
quilisma	ql qi		ql- qi-				ql> qi>	
pes stratus	pt							

Table from Cardine's Gregorian Semiology, pp. 12-13 with nabc strings and gregall glyphs.

Name of Symbols	Simple Symbols	Symbols differentiated by			Symbols indicating a special meaning		
		the modification		the addition of letters	melodic	phonetic	
		of the mark	of the grouping (neumatic break)			Liquescence	
						augment.	diminut.
uncinus	un						
punctum	pu						
virga	vi	viS vipp2su2vihh				vi>1	
tractulus	ta	ds ts		talst2 tslsc2lst2		ta>2	
clivis	cl	clG		cllsn9 cllst9 clGlsa5	clG1 clM	cl> cl>1	vi> ta>2
pes	pe	peG		pelsn3 pelst2 pelsc2		pe>1	ta>
porrectus	po	poG			poM	po>	
torculus	to	toS1	toG	toGlsa5 toSlsa5 toSsun1lsa5 tolst9		to>	
climacus	ci	pusu1sun1 visu2 unsun2		visu2lst3			cl>1
scandicus	sc sc1	vippn1pp1 vippn2	pevihl	sclst2 pelst2vihl pelsa2vihl		vi>1pp2	ta>ppn1 ta>pp2
porrectus flexus	pf1	pfS	unhg`to po`un	unhglst2`to po`unlsa2	pfM	pf>	
(pes subbipunctis)	tosu1	tosun1	vippn1su2 unppn1sun2 visu1sun1	tosu1lsc5 tosun1lsa5 unlst2`puhisu1sun1			
scandicus flexus	sf		unhd`to sfG	unhdlst2`to pelst2clhh			
torculus resupinus	tr	trS	poppn1 pe`peG tovihg toGvihg	trSlsc1 trlst3 trlst9		tr>	pe`ta>
clivis + pes or torculus	cl!pe cl!to	→ toS!pe	toS!to				
oriscus	or or1	isolated & at the syllabic articulation; inside the neume	ql oc oc!vi pq pq1 tg cl1!or cl1!pi cl1!pi!vi cl1!pi!cl cl1!pi!to to!or sa sa1 vs vsG			or>	
quilisma	ql ql!cl ql!po						
pes stratus		pt1					

Table from Cardine's Gregorian Semiology, pp. 14-15 with nabc strings and grelaon glyphs.

Table with nabc strings and corresponding gregall and gresgmodern glyphs:

bv		
bv-		
bv-sut2		missing
bv>		
ci		
ci-		
ci-pp2		
ci1		
ci>		
ciG		
ciG1		
ciM		
ciM-	missing	
ciS		
ciS1		
ciSppt1		
cilsc3		missing
cippt1		
cippt1lsc3		missing
cippu1		
cippu1lsc3		missing
ci~		
cl		
cl!cl>1		
cl!clS		
cl!clSsut2		
cl!pi		
cl!pilsc1lst3		missing
cl!pilsc2		missing
cl!po		
cl!po-1		
cl!po>		
cl!po>ppt1		
cl!pr		
cl-		
cl-!pi		
cl-1su2		
cl->		
cl->1		
cl-lst2		missing
cl-ppt1	missing	
cl-ppt2		
cl>		
cl>1		
cl>pp2	missing	
cl>ppt2		
clM		
clM-		
clMpp2	missing	
clS		
clS-		
clS-1		
clS-su2		
clS1		
clSsu1sut1		
clSsu2		
clSsut1		
clSsut1sux1	missing	
clSsut2		
clSsuu1su2		missing
clSsux1	missing	
cllsc3		missing
cllst2		missing
clpp2		
clppt1		
clppt2		
cl~		
ds		
ds-		
ds>		
dslsx6		missing
gr		
gr-		
gr-lsc6		missing
lsal		
lsam		
lsb		
lsc		
lscm		
lsco		
lscw		missing
lsd		missing
lse		
lseq		missing
lsew		missing
lsfid		missing
lsfr		
lsg		
lsi		
lsim		
lsiv		missing
lsk		
lsl		
lslb		
lslc		
lslen		missing
lslm		
lslp		
lslt		
lsm		
lsmoll		
lsp		
lspar		missing
lspfec		missing
lspm		
lspulcre		missing
lss		
lssb		
lssc		
lssimil		
lssimul		missing
lssm		
lsst		
lssta		missing
lst		
lstb		
lstm		
lstw		missing
lsv		missing
lsvol		missing
lsx		
or		
or>		
pe		
pe!vi		
pe-		
pe-1		
pe-1su1sut1		
pe-1su1sut2		
pe-1su1suu1		
pe-1su1suu2		
pe-1su1suw1		
pe-1su2		
pe-1sut1suw1		missing
pe-1sut1sux1		
pe-1sut2		
pe-1sut3		
pe-1suu1su2		missing
pe-1suw2		
pe->		
pe->1		
pe-su1sut1		
pe-su2		
pe-su2suy1		
pe-su3sut1		
pe-su4		
pe>		
pe>1		
pe>2		
pe>ppt1	missing	
peG		
peS		
peSsu1sux1		
peSsu2		
peSsut1su2		
peSsut2		
peSsut3		
peppt1		
pesu1sut1		
pesu1sut2		missing
pesu1sut3		
pesu1suu1		
pesu1suw1		
pesu1sux1		
pesu2		
pesu2sut1		missing
pesu2suv1		
pesu3		
pesut2		
pesuv1su1sut1		
pe~		
pf		
pf!cl-		
pf!clM-		
pf!clM-ppt1		
pf!cl~		
pf!cl~ppt1		
pf!pr		
pf-		
pf1lsc2		missing
pf>		
pfG		
pfM		
pfM-	missing	
pfMsut1		
pfMsuu1	missing	

pfS		
pflsc2		missing
pflst2		missing
pf~		
pi		
pi>		
piM	missing	
pilsc2		missing
pilst2		missing
pisu1		
pisut1		
po		
po-		
po-1		
po-1pp2su1sux1		
po-1su1sux1		
po-1su2		
po-2	missing	
po>		
poG		
poG-		
poG-1	missing	
poG1	missing	
polsc2		missing
posu1sut1		missing
posu2		
po~		
pp1		
pp2		
pp3		
pp4		
pp5	missing	
ppt1		
ppt2		
ppt3		
ppt4		
ppt5		
ppu1		
ppu2		
ppu3		
ppu4		
ppu5		
pq		
pq!cl		
pq!cl-		
pq!cl>		
pq-		
pq-su2		
pq>		
pqS		
pqS-	missing	
pqS>		
pqlsfr1		missing
pr		
pr-		
pr-lsc3		missing
pr>		
prM		
prlsc2		missing
prlst2		missing
pr~		

pt		
pu		
qi		
qi!cl		
qi!cl-		
qi!cl>		
qi!clppt1		
qi!po		
qi!po>		missing
qi!vs		
qi!vsppt1		
qi!vsppt1su1sut2		missing
qi!vsppt1su2		
qi!vssu1sut2		missing
qi!vssu2		
qi-		
qi-ppt1su1sut2		
qi-ppt1suu1su1sut2		missing
qi-su1sut2		
qi-su1sux1		
qi-su2		
qi-sut2		
qi-sut3		
qi-suu1su1sut2		missing
qi>		
qi>ppt1		
qippt1		
qippt2sut2		
qisu2		missing
qisut1sux1		
qisut2		
ql		
ql!cl		
ql!cl!cl>		
ql!cl!cl>ppt1		
ql!cl!pi		
ql!cl!pippt1		
ql!cl-		
ql!cl->		
ql!cl->ppt1		
ql!cl-ppt1		
ql!cl>		
ql!cl>ppt1		
ql!clM		
ql!clMppt1		
ql!clS	missing	
ql!clS-2		
ql!clS-2ppt1		
ql!clS-2ppt1sut1		
ql!clS-2sut1		
ql!clppt1		
ql!pf		
ql!pf-		
ql!po		
ql!po-1		
ql!po-1ppt1		
ql!po>		
ql!poppt1		
ql!po~		
ql!po~ppt1		
ql!vs		
ql!vsppt1		
ql!vsppt1su2		

ql!vsppt1sut2		
ql!vssu2		
ql!vssu3		
ql!vssut2		
ql-		
ql-ppt1		
ql-ppt1su1sut2		
ql-ppt1su1suu1		
ql-ppt1su1sux1		
ql-ppt1su2		
ql-ppt1su3		
ql-ppt1sut2suu1		
ql-ppt1suu2		
ql-ppt2su1suv1su2		
ql-su1sut1		
ql-su1sut2		
ql-su1suu1		
ql-su1suu2		
ql-su1suv1su2		
ql-su1sux1		
ql-su2		
ql-su3		
ql-sut1sux1		
ql-sut2		
ql-sut2suu1		
ql-suu2		
ql>		
ql>ppt1		
qlppt1		
qlppt1su2		missing
qlppt1sut3		
qlsu2		missing
qlsut2		
qlsut3		
sa		
sa!cl		
sa!cl-		
sa!cl>		
sa!clS-1		
sa!clS1		
sa!clS1lsc2		missing
sa-		
sa-1		
sa->1		
sa-lsfr1		missing
sa-su2sut1		
sa1		
sa1!cl->		
sa1!cl>		
sa>		
sa>1	missing	
sa>2		
saG		
saM		
saM-1		
saM-4		
saM1		
saM2		
saM3		
saM4		

saM5 missing
saM5ppt1 missing
salsfr1 missing
sapp1
sa~
sc
sc-
sc-1 missing
sc1
sc>
scG
scG-1
scG1
scM missing
scM1
scM2
scS
sc~
sf
sf-
sf> missing
sfG missing
sfG1 missing
sfM
sfM1 missing
sfS
sflsc3 missing
st
st-
st>
stS
su1
su1sut1
su1sut1su1sut1
su1sut1su2
su1sut1sux1
su1sut1suy1
su1sut2
su1sut2su1
su1sut3
su1suu1
su1suu1su1
su1suu1su2
su1suu2
su1suv1
su1suv1su2
su1suw1
su1sux1
su2
su2sut1
su2suv1
su2suw1
su2sux1
su2suy1
su3
su3sut1
su4
su5
sut1
sut1su1sux1 missing
sut1su2
sut1suu1
sut1suw1 missing
sut1sux1
sut2
sut2su2
sut2suu1
sut2sux1 missing
sut2suy1
sut3
sut3suw1
sut3suy1
sut4
suu1
suu1su1sut2 missing
suu1su2
suu2
suu2su2
suu3
suv1
suv1su1sut1
suv1su2
suw2
ta
ta-
ta>
ta>ppt1
talsc3 missing
talst2 missing
tg
tg1
tgM
tgS
tgS-
tgS1
tgS2 missing
tgS3
tgS4
tgSsuu1
tgsuu1
to
to!ciG
to!ciGhh missing
to!cl
to!cl!cl
to!cl!cl!cl
to!cl!pi
to!cl!po
to!cl!po-1
to!cl-
to!cl-!cl-
to!cl-1
to!clS
to!pi
to!po
to!posu2
to!pr
to-
to-1
to-1su1
to-1su2
to-1suu1 missing
to-1suu1su2
to->
to>
to>1
to>2 missing
to>pp2
toG
toM
toM- missing
toM1
toM2 missing
toM2- missing
toM3
toS
toS1
toS2
toS2su2
toS2sut1
toS2sut2
toS2sux1
toS3
toS3lst2 missing
toS4 missing
toSsu2
tolsc3 missing
tolsc4lst3 missing
tolst2 missing
toppt1 missing
toppu1
tr
tr- missing
tr-1
tr-1su2
tr-2 missing
tr-3
tr-~
tr-~1
tr>
trG
trG1
trG2
trM
trS
trS-
trS1
trSsut2
trlst2 missing
trsu3 missing
tr~
ts
ts-
ts>
tsM
tsM-
tslsx6 missing
tv
tv-
vi
vi-
vi-!ci
vi-!vi!vi
vi-!vi-!vi
vi->
vi-lst3 missing
vi-pp2
vi-pp4
vi-ppt2
vi-ppt3

vi-su1sut1		
vi-su1sut1su2		
vi-su1sut2		
vi-su1sut2su1		
vi-su1suu1		
vi-su1suu1su1		
vi-su1suu1su2		
vi-su1suu2		
vi-su1suw1	missing	
vi-su1sux1		
vi-su2		
vi-su2sut1		
vi-su2suw1	missing	
vi-su2sux1		
vi-su3		
vi-su4		
vi-sut1su2		
vi-sut1suu1		
vi-sut1sux1	missing	
vi-sut2		
vi-sut2sux1	missing	
vi-sut3		
vi-sut4		
vi-suu1su2		
vi-suu2		
vi-suu2su2		
vi-suu3		
vi>		

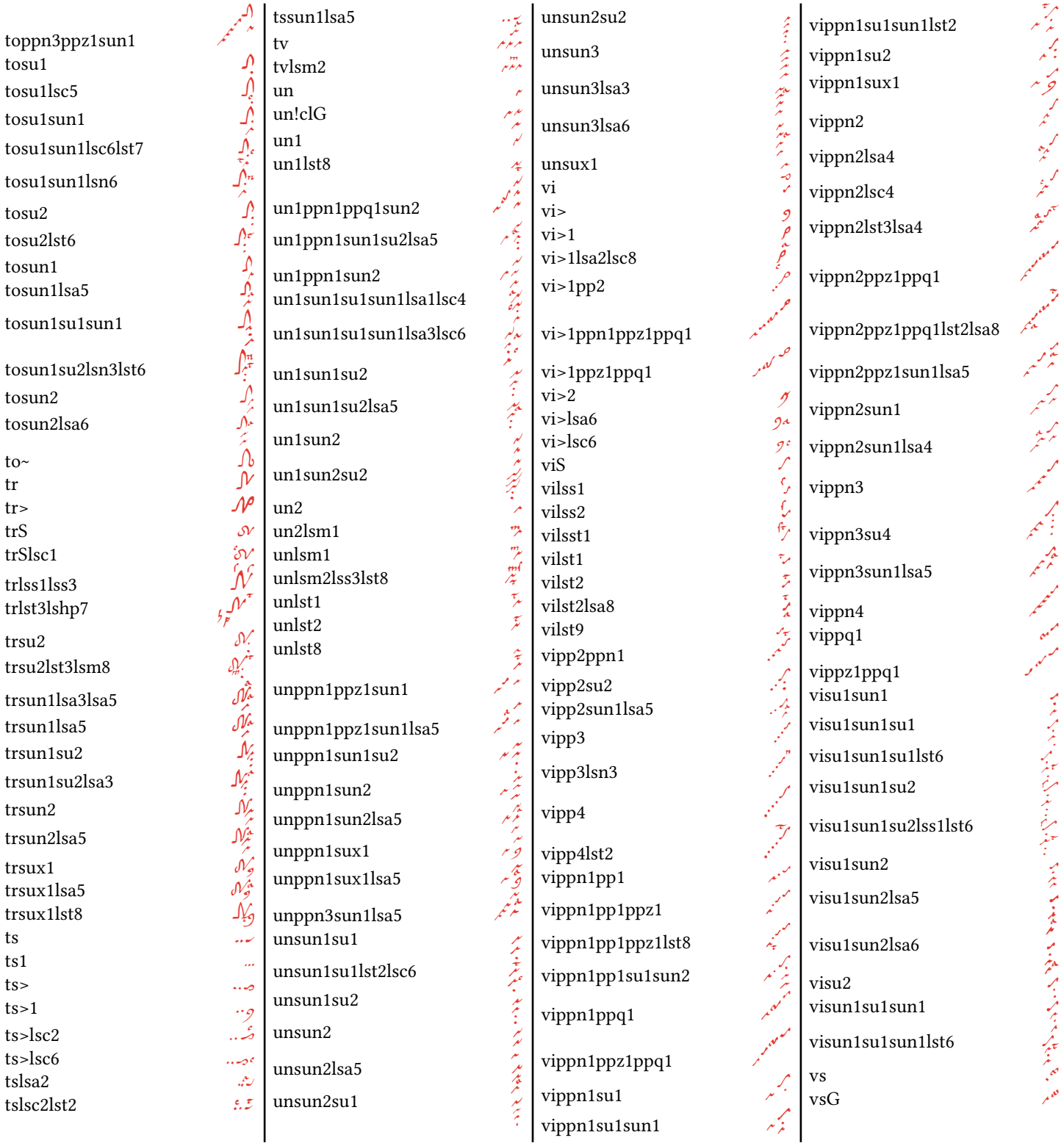
vi>pp2		
vi>ppt2		
vi>ppt3		
vilsc3		missing
vilst1		missing
vipp2		
vipp3		
vipp4		
vippt1su1sut1		
vippt1su2		
vippt1su2lsc3		missing
vippt2		
vippt3		
vippu1su2		
vippu1su2lsc3		missing
visu1sut1		
visu1sut1su1sut1		
visu1sut1su2		
visu1sut1sux1		
visu1sut1suy1		
visu1sut2		
visu1suu1		
visu1suu1su2		
visu1suv1		
visu1suw1		
visu1sux1		

visu2		
visu2lsc3		missing
visu2sut1	missing	
visu2suw1		
visu2sux1		
visu3		
visu5		
visut1su2		
visut1sux1		
visut2		
visut2su2		
visut2sux1	missing	
visut2suy1		
visut3		
visut3suw1		
visut3suy1		
visut4		
visuu1su2		
visuu2su2		
visuv1su2		
vs		
vs-		
vs->		
vs>		
vslsc3		missing
vslst2		missing
vs~		

Table with nabc strings and corresponding grelaon glyphs:

bv
bvlsa2
bvsun4
ci
cilsf1
cilsn8
cl
cl!cl!cl
cl!cl!cllsn9
cl!cl!clsun1lsa5
cl!cl!pi
cl!cl!pi!vi
cl!cl!po
cl!cl!to
cl!or
cl!orpp2
cl!pe
cl!pelsc2
cl!pelsn7
cl!pelst2
cl!pi!vi>
cl!po
cl!popp3
cl!to
cl!to!cl!cl
cl!to!pi
cl!to!pi!cl
cl!to!pi!clpp2
cl!to!pi!clpp2lsn9
cl!to!pipp2
cl!to>
cl!tolst9
cl!topp2
cl!toppz1su1sun1
cl!tosu1
cl!tosu1lsc6
cl!tosu1lsc6lst8
cl!tosun1lsa5
cl1!or
cl1!orlsc3
cl1!orlss3
cl1!orpp2
cl1!orsun1
cl1!orsun1lsa5
cl1!pi
cl1!pi!cl
cl1!pi!orsun1
cl1!pi!pesu2
cl1!pi!pi!vi
cl1!pi!pisun1lsa5
cl1!pi!to
cl1!pi!tosu1
cl1!pi!tosun1
cl1!pi!vi
cl1!pi>
cl1!pilsm3lsa6
cl1!pilsn8
cl1!pilsn9
cl1!pilst9
cl1!pipp2
cl>
cl>1
cl>1pp2
cl>lsa5
clG
clG1
clGlsa5
clGlsnl2
clM
cllshn8
cllsm2
cllsn8
cllsn9
cllsnt2
cllsst6
cllsst8
cllst9
clsu2sun1lst6
clsu3lst5
clsun1lst5
ds
ds1
ds1!clsu1sun1
ds1!clsu1sun1lst5
ds1!clsun2
ds1!clsun2lst5
ds1!ds1
ds1!ts
ds1!ts1
ds1pp3
dslst2
dslst8
lsa
lsc
lseq
lseq-
lsequ
lsf
lsh
lshn
lshp
lsl
lsm
lsmd
lsn
lsnl
lsnt
lss
lssimp
lssimpl
lssp
lsst
lst
lsth
ltdo
ltdr
ltdx
lti
ltps
ltqm
ltsb
ltse
ltsj
ltsl
ltsn
ltsp
ltsr
ltst
ltus
oc
oc!cl
oc!clM
oc!pe
oc!pe!pu
oc!vi
oc!visun1
ocG
ocG1
ocGlsa5
oclst8
or
or1
or>
or>lsa1
orlst3
orsu1sun1
orsu1sun1lsc2
orsu1sun1lsc3
pe
pe!or
pe!pu
pe!ta
pe!ta>2
pe!ta>2lsa2
pe!talsa2
pe!talsc1lst2
pe!un
pe!unlsc2
pe>
pe>1
pe>1lsa6
peG
peG1
peGlsa6
peGlsf1
pelsa2
pelsc2
pelsl2
pelsm2
pelsn2
pelsn3
pelsst1lsm4
pelsst2
pelst2
pesu2
pesu2lst3
pesu2sun1
pesun3
pesun3lss1lsa3
pf
pf1
pf2
pf>
pf>1
pf>2
pfM
pfMlsmd3
pfS
pfppn1ppq1sun1lsa5
pfsun1
pfsun1lsn5
pf~
pf~1
pi
po
po>
po>1
po>pp2
po>ppn1
po>ppn1ppz1
poG
poM
poMsun1
polsf2lst3
polst2
popp2
popp2su1sux1
poppn1
posu1sun1
posun1lsa5
posux1
posux1lsa5
pp1
pp2
pp2ppn1
pp3

pp4
ppn1
ppn1pp1
ppn1pp1ppz1
ppn1ppq1
ppn1ppz1
ppn1ppz1ppq1
ppn2
ppn2ppz1
ppn2ppz1ppq1
ppn3
ppn3ppz1
ppn4
ppq1
ppz1
ppz1ppq1
pq
pq1
pr
pt1
pt1ppn1
pt2
pt3
pu
pu1
pulsnt3
pupp1su1sun1
pupp1su2
pupp2su1sun1
pupp2su2
puppn1su1sun2
puppz1su1sun2
puppz1su2
pusu1sun1
pusu1sun1lsf1
pusu1sun1lst7
pusu1sun2
pusu1sun2lsa5
pusu2sun1
pusu3
pusu4
ql
ql!cl
ql!cl!or
ql!cl!orppn1ppz1
ql!cl!to
ql!cl!to!or
ql!cl!to!orsun1
ql!cl!tolst6
ql!cl!toppn1
ql!cl!tosu3
ql!cl!tosu3lst6
ql!cl!tr>
ql!cl!tr>ppz1
ql!clppn1
ql!po
ql!po>
ql!po>ppn1
ql!polsa1lsa3lsa9
ql!polsm1
ql!poppn1
ql!poppn1lst2
ql!poppn1sun1
ql!poppn1sux1
ql!poppz1
ql!poppz1lsa3
ql!poppz1su1sun1
ql!posu1sun1
ql!posun1
ql!posux1
ql>
ql>ppn1
ql>ppn1lsc3
qllsst2
qlppn1
qlppn1ppz1
qlppn1ppz1lst2
qlppn1ppz1su1sun1
qlppn1ppz1sun1
qlppn1ppz1sun1lsa5
qlppn1su1sun1
qlppn1su1sun2
qlppn1su2
qlppn1su2lsnl1
qlppn1sun1
qlppn1sun1lsc5
qlppn1sun1su2
qlppn2sun1
qlppn2sun1lsa5
qlppn2sun2
qlppn2sun2lsa5lsa5
qlppz1su1sun1
qlppz1su2
qlsu1sun1
qlsu1sux1
qlsu2
qlsun1
qlsun1lsa5
qlsun1su2
qlsun2
qlsun2lsa5
qlsux1
sa
sa1
sa1lsc4
sa1lsc6
sa1sun1lsa5
sa>
salsc4
salsc6
sc
sc1
sc1lsc4
sclst2
sf
sfG
su1
su1sun1
su1sun1su1
su1sun1su2
su1sun2
su1sux1
su2
su2sun1
su3
su4
sun1
sun1su1
sun1su1sun1
sun1su2
sun2
sun2su1
sun2su2
sun3
sun4
sux1
ta
ta>
ta>1
ta>2
ta>2lsc2
ta>lsnl2
ta>pp2
ta>ppn1
tasu2
tg
tg!cl
tg!cl>1
tg!cllsn9
tg!or
tglsn8
tglst9
tgsun1lsa5
to
to!cl
to!cl!cl
to!cl!cllst9
to!cl!clsu1
to!cl!clsun2
to!cl!clsun2lsa5
to!cl!orsun1
to!cl!pi
to!cl>1
to!clM
to!clsu1sun1
to!clsu1sun1lsm6
to!clsu2
to!clsu2lsn6
to!clsun1
to!clsun1lsa5
to!or
to!orlsc6
to!orlst3
to!orsun1lsa5
to!pi!cl
to!pi!cl!pi!orsun1lsa5
to!pi!cl!pi!pi
to!po
to!pohd
to!to
to!to!pi!cl
to!to!pi!clpp2
to!toppn1sun1
to!tosun1
to>
to>2
to>2lsa3
to>2lsc1lsa3
to>lsa5
to>pp2
toG
toGlsa1lst8
toGlsa5
toS
toS!pe
toS!to
toS!tolst9
toS1
toSlsa5
toSsun1lsa5
tolsf4
tolsn8
tolsnt2lst9
tolst9
tolsth8
topp1su1
topp1su1lsa6
topp2
topp2sun1


toppn3ppz1sun1
tosu1
tosu1lsc5
tosu1sun1
tosu1sun1lsc6lst7
tosu1sun1lsn6
tosu2
tosu2lst6
tosun1
tosun1lsa5
tosun1su1sun1
tosun1su2lsn3lst6
tosun2
tosun2lsa6
to~
tr
tr>
trS
trSlsc1
trlss1lss3
trlst3lshp7
trsu2
trsu2lst3lsm8
trsun1lsa3lsa5
trsun1lsa5
trsun1su2
trsun1su2lsa3
trsun2
trsun2lsa5
trsux1
trsux1lsa5
trsux1lst8
ts
ts1
ts>
ts>1
ts>lsc2
ts>lsc6
tslsa2
tslsc2lst2
tssun1lsa5
tv
tvlsm2
un
un!clG
un1
un1lst8
un1ppn1ppq1sun2
un1ppn1sun1su2lsa5
un1ppn1sun2
un1sun1su1sun1lsa1lsc4
un1sun1su1sun1lsa3lsc6
un1sun1su2
un1sun1su2lsa5
un1sun2
un1sun2su2
un2
un2lsm1
unlsm1
unlsm2lss3lst8
unlst1
unlst2
unlst8
unppn1ppz1sun1
unppn1ppz1sun1lsa5
unppn1sun1su2
unppn1sun2
unppn1sun2lsa5
unppn1sux1
unppn1sux1lsa5
unppn3sun1lsa5
unsun1su1
unsun1su1lst2lsc6
unsun1su2
unsun2
unsun2lsa5
unsun2su1
unsun2su2
unsun3
unsun3lsa3
unsun3lsa6
unsux1
vi
vi>
vi>1
vi>1lsa2lsc8
vi>1pp2
vi>1ppn1ppz1ppq1
vi>1ppz1ppq1
vi>2
vi>lsa6
vi>lsc6
viS
vilss1
vilss2
vilst1
vilst2
vilst2lsa8
vilst9
vipp2ppn1
vipp2su2
vipp2sun1lsa5
vipp3
vipp3lsn3
vipp4
vipp4lst2
vippn1pp1
vippn1pp1ppz1
vippn1pp1ppz1lst8
vippn1pp1su1sun2
vippn1ppq1
vippn1ppz1ppq1
vippn1su1
vippn1su1sun1
vippn1su1sun1lst2
vippn1su2
vippn1sux1
vippn2
vippn2lsa4
vippn2lsc4
vippn2lst3lsa4
vippn2ppz1ppq1
vippn2ppz1ppq1lst2lsa8
vippn2ppz1sun1lsa5
vippn2sun1
vippn2sun1lsa4
vippn3
vippn3su4
vippn3sun1lsa5
vippn4
vippq1
vippz1ppq1
visu1sun1
visu1sun1su1
visu1sun1su1lst6
visu1sun1su2
visu1sun1su2lss1lst6
visu1sun2
visu1sun2lsa5
visu1sun2lsa6
visu2
visun1su1sun1
visun1su1sun1lst6
vs
vsG

manuscript sources of the gregall font

The list of glyphs uses following abbreviations for manuscript references; the page number is always for the start of some antiphon, responsory, mass propers, even when the actual neume is on a later page. At the end of lines, GT stands for Graduale Triplex, Solesmes, 1979, OT for Offertoriale Triplex, Solesmes, 1985, AM for Antiphonale Monasticum, Solesmes, 1934, ARII for Antiphonale Romanum II, Solesmes, 2009 and NR for Nocturnale Romanum, Hartker Verlag, 2002, followed by page number which also contains the corresponding propers, antiphons or responsories.

Most of the glyphs taken from the e-codices Virtual Manuscript Library of Switzerland, http://www.e-codices.unifr.ch/en, with permission from Stiftsbibliothek, St. Gallen http://www.stibi.ch/ and Stiftsbibliothek, Einsiedeln http://kloster-einsiedeln.ch/, the picture fragments from there are only present in the font source file to help drawing the glyphs.

Cantatorium codex 359
C1 = http://www.e-codices.unifr.ch/en/download/csg-0359_001/max
C25U Universi GT16
C26A Alleluia Ostende GT16
C26D Dilexisti iustitiam GT499
C27E Ex Sion GT18
C27Q Qui sedes GT22
C28A Alleluia. Excita Domine GT23
C28T Tollite portas GT25
C32B Benedictus es in firmamento GT664
C35Q Qui regis Israel intende
C37A Alleluia. Veni, Domine GT36
C38T Tecum principium GT42
C38A Alleluia Dominus dixit GT43
C39B Benedictus qui GT45
C40A Alleluia Dies sanctificatus GT49
C40V Viderunt omnes GT48
C42A Alleluia. Hic est discipulus GT636
C43A Anima nostra GT453
C43L Laus tibi Christe
C44E Ecce sacerdos GT486
C44D Diffusa est gratia GT408
C45S Speciosus forma GT54
C46O Omnes de Saba GT57
C46B Benedictus Dominus GT257
C46A Alleluia Vidimus stellam GT58
C47I Iuravit Dominus GT486
C49I Inveni David GT445
C50G Gloriosus Deus GT456
C52T Timebunt gentes GT265
C52P Posuisti Domine GT477
C53S Suscepimus Deus GT360
C53A Alleluia. Adorabo ad templum GT270
C54A Audi filia GT417
C56B Beatus vir GT475
C56D Desiderium animae GT432
C57B Beatus vir GT481
C58A Adiutor GT69
C59D De profundis GT673
C60S Sciant gentes GT88
C60C Commovisti Domine GT89
C61T Tu es Deus GT275
C61I Iubilate Domino GT185
C62M Miserere mei GT63
C64Q Qui habitat GT73
C67D Dirigatur GT340
C71L Laudate Dominum GT187
C73P Propitius esto GT288
C74B Bonum est GT327
C78O Oculi omnium GT343
C79L Leatatus sum GT336
C80Q Qui confidunt GT109
C81E Exsurge Domine GT114
C82B Beata gens GT333
C82R Respice Domine GT320
C82V Venite filii GT298
C84T Tibi Domine GT118
C87E Exaltabo te GT112
C88T Tollite hostias GT272
C89T Tenuisti manum GT133
C95D Domine exaudi GT172
C97D Domine audivi
C100C Crux fidelis GT182
C103C Cantemus Domino GT186
C104V Vinea facta GT188
C107A Alleluia Pascha nostrum GT197
C118C Clamaverunt iusti GT454
C122I In omnem terram GT427
C123C Constitues eos GT426
C132D Domine praevenisti GT509
C134B Benedicite Dominum GT608
C137A Audi filia GT406
C139A Alleluia Benedictus es GT375
C139B Benedictus es GT372
C145L Liberasti nos GT366
C148A Alleluia Domine refugium GT321
C148Q Alleluia Quoniam Deus GT327
C148V Alleluia Venite GT324
C149A Alleluia. Paratum cor meum GT344
C150L Alleluia Laetatus GT19
C153S Alleluia Sancti tui, Domine, benedicent GT463
C153B Alleluia Beatus vir qui suffert GT511
C154A Alleluia Gaudete iusti GT430

Einsiedeln codex 121
E1 = http://www.e-codices.unifr.ch/en/download/sbe-0121_001/max
E2R Respice in me OT5
E2D Dominus dabit GT17
E3D Deus tu convertens GT20
E10P Prope est Dominus GT35
E11C Confortamini OT9
E13P Prope es tu GT24
E16B Benedictus es
E20E Exsulta satis GT28
E23T Tollite portas OT14
E26L Lux fulgebit GT44
E29E Exsulta filia sion GT47
E31T Tui sunt caeli OT18
E33V Viderunt omnes GT50
E34I In virtute tua OT152
E35E Elegerunt apostoli OT161
E39E Exiit sermo inter GT636
E42A Anima nostra GT466
E51R Reges tharsis OT21
E64L Laetamini in Domino OT140
E69P Posuisti Domine GT477
E75G Gaudeamus omnes GT545
E82D De profundis GT673
E92I Immutemur GT65
E104S Sicut oculi GT77
E125P Precatus est Moyses GT317
E130I Illumina oculos GT290
E131O Oculi mei semper GT96
E133I Iustitiae Domini GT309
E147L Laetare Ierusalem GT108
E148L Laudate Dominum GT110
E155B Benedicite gentes Domino GT231
E158D Domine ad adiuvandum OT42
E161S Sitientes venite GT114
E162F Factus est Dominus GT119
E166C Confitebor tibi Domine OT44
E173E Eripe me GT129
E193D Domine, exaudi orationem OT53
E195D Dominus Iesus GT165
E205R Resurrexi GT196
E213C Alleluia Christus resurgens GT226
E248V Viri Galilaei GT235
E255S Spiritus Domini GT252
E256C Confirma hoc OT79
E262S Sapientiam sanctorum GT452
E272M Mihi autem nimis OT128
E293V Vindica Domine
E320I Inclina aurem GT300
E327O Oravi Deum meum OT107
E324R Respice, Domine GT320
E336I In voluntate tua GT346
E339S Si iniquitates GT350
E373B Benedictus es GT372

Hartker codex 390/391

H1 = http://www.e-codices.unifr.ch/en/download/csg-0390_001/max
H195 = http://www.e-codices.unifr.ch/en/download/csg-0391_001/max
H16A Aspiciebam in visu NR111
H16M Missus est Gabriel NR111
H16V Ave Maria NR113
H22S Sicut mater consolator NR127
H23E Egredietur Dominus de Samaria NR129
H25E Ecce mitto Angelum AM201
H26G Grata facta est NR25*
H28Q Qui venturus est NR134
H29E Ecce radix Jesse NR137
H32N Nascetur nobis NR149
H33I Iuravi dicit Dominus NR151
H34F Festina ne tardaveris NR130
H35G Germinaverunt campi NR144
H36A Adnuntiatum est
H48B Beata et venerabilis
H50Q Quem vidistis pastores AM240
H72V Venit lumen tuum AM290
H72H Hodie in Iordane NR229
H73O Omnes de Saba NR237
H73S Stella ista NR242
H74I Illuminare, illuminare NR236
H75I In columbae specie NR231
H75V Videntes stellam NR242
H77S Stella ista AM291
H77A Ab Oriente AM297
H87E Exaudi Deus
H103S Spes nostra salus NR512
H103M Magnus Dominus NR516
H104T Te Deum patrem
H112S Stans a dextris AM798
H114C Corde et animo AM1033
H114N Nativitas tua AM1035
H118G Gaude, Maria Virgo NR130*
H119C Cum inducerent
H173I Insurrexerunt in me NR364
H173C Contumelias et terrores NR365
H179L Liberavit Dominus ARII183
H179A Amicus meus osculi NR381
H179I Iudas mercator NR382
H179U Unus ex discipulis NR383
H180U Una hora non potuistis NR388
H219T Tradiderunt me NR408
H219I Iesum tradidit impius NR409
H225M Mulieres sedentes AM449
H266D Domino si in tempore
H269S Spiritus Sanctus
H269D Dum complerentur NR493
H269R Repleti sunt omnes NR494
H270S Spiritus Domini AM520
H271L Loquebantur variis AM521
H297B Beatam me dicent NR[172]
H305H Hodie nata est NR278*
H305G Gloriosae virginis Mariae NR279*
H306N Nativitas gloriosae NR280*
H307N Nativitas tua NR281*
H307F Felix namque NR[171]
H320R Regem confessorum
H322B Beatus Gallus zelo
H322C Columbanus itaque beato
H323B Beatus Gallus cum
H323D Domine Iesu Christe
H324P Pater sanctus
H324V Vir Deo plenus
H325I Iste sanctus digne
H367V Viri sancti NR[54]
H390C Credo quod Redemptor NR[186]
H392L Libera me Domine de viis NR[200]
H392I Libera me Domine de morte NR[201]
H424M Magnificat
H428A Ascendens Iesus AM593
H443V Venite exsultemus NRXXVIII

St. Gall codex 339

G1 = http://www.e-codices.unifr.ch/en/download/csg-0339_033/max
G56 = http://www.e-codices.unifr.ch/en/download/csg-0339_088/max
G57 = http://www.e-codices.unifr.ch/en/download/csg-0339_088a/max
G58 = http://www.e-codices.unifr.ch/en/download/csg-0339_089/max
G8E Exulta satis OT11
G9A Alleluia Veni Domine GT36
G11P Puer natus est GT47
G11V Viderunt omnes GT48
G12T Tui sunt caeli OT18
G20I Iubilate Deo universa terra GT227
G22D Domine quinque talenta GT515
G25P Posuisti Domine GT477
G30E Exsurge quare obdormis GT91
G35D Domine vivifica me OT31
G42D Domine Deus salutis OT112
G44P Precatus est Moyses GT317
G45D Domine in auxilium OT106
G51I Intende voci GT280
G51S Si ambulabem GT125
G57F Factus est Dominus GT119
G89D Domine Deus in simplicitate OT159
G113A Alleluia. Dilexit Andream GT625
G116C Cantabo Domino GT283
G126D Alleluia. Domine Deus meus GT282
G128A Alleluia. Attendite populе
G132A Alleluia. Mirabilis Dominus GT462
G132V Alleluia. Vox exultationis
G139A Asperges me Domine
G140O Oremus dilectissimi nobis
G141O Omnipotens Deus

St. Gall codex 376

N1 = http://www.e-codices.unifr.ch/en/download/csg-0376_001/max
N39H Hodie cantandus est nobis
N53A Audite fratres dilecti
N59N Nativitatem venerandam
N65G Gloriatis laus tua Deus
N67G Gloria in excelsis Deo
N68G Gloria in excelsis Deo
N69G Gloria in excelsis Deo
N73G Kyrie eleison
N86D Dilexisti iustitiam GT498
N100T Tui sunt caeli OT18
N171D Deus exaudi orationem GT107
N221N Alleluia. Non vos relinquam GT242
N270A Alleluia. Benedictus es GT375
N293A Alleluia. Sancti tui, Domine, benedicent GT463
N296V Alleluia. Veni Sancte Spiritus GT263
N363C Carmen suo dilecto
N368L Letamente canamus Domino nostro
N372O O quam mira sunt Deus

St. Gall codex 374

A1 = http://www.e-codices.unifr.ch/en/download/csg-0374_001/max
A17T Tui sunt caeli OT18
A22V Vox in Rama GT638
A26F Fili quid fecisti GT51
A56C Custodi me Domine GT304
A127E Exultabunt sancti OT143
A167O Oravi Deum meum OT107
A174A Alleluia. Eripe me GT308

Bamberg, lit. 6, Gradual of St. Emmeram

B9v = http://bsbsbb.bsb.lrz-muenchen.de/~db/0000/sbb00000128/images/index.html?id=00000128no=19&seite=22&signatur=Msc.Lit.6
B9vA Anima nostra GT466

	virga			
	vi	C78O	tu	
	vilsc3	C82R	testamentum	
	vilst1	E29E	rex	2.166x
	vi-	C80Q	montes	
	vi>	C27E	eius	
	vi-lst3	H50Q	Quem	
	vi->	E20E	sanctus	2.166x
	tractulus			
	ta	C26A	Domine	
	talsc3	E262S	saeculum	2.166x
	talst2	C56D	Desiderium	
	ta-	C81E	antiquis.	
	ta>	C25U	et	
	suv1 *Cf.*	pesu2suv1		
	punctum			
	pu	C137A	Audi	
Cf.	st			
Cf.	ci>			
	gravis			
	gr	H271L	Loquebantur	
	gr-	H270S	Spiritus	
	gr-lsc6	H270S	replevit	
	clivis			
	cl	C78O	sperant	
	cllsc3	C26A	nobis	
	cllst2	C25U	exspectant	
	cl-lst2	C52P	pretioso.	
	cl-	C78O	omnium	
	clS-	C81E	opem	
	clS	C78O	opportuno.	
	clS1	H424M	potentiam	
	clM	C148V	nostro	
Cf.	cl>			
	cl>1	H36A	virginis	
Cf.	vi>			
	cl->1	H50Q	natum	
	cl->	H50Q	angelorum	
	clM-	C27Q	super	
	clS-1	H428A	navem	
	pes			
	pe	C80Q	Domine	
	pe-	C80Q	aeternum	
	pe-1	C64Q	Scapulis	
	peS	C78O	in	
	peG	B9vA	erepta	0.5x
Cf.	peG			
	pe>	C100C	germine	
	pe>1	C27E	eius	
	pe>2	C71L	misericordia	
	pe->	N296V	tuorum	
Cf.	ta>			
	pe->1	H35G	Germinaverunt	1.7x
	porrectus			
	po	C139B	super	
	polsc2	C82R	iudica	
	po-	G20I	anime	0.85x
	po-1	C26A	tuum	
	poG	C40V	salutare	
	po>	C80Q	commovebitur	
	po~	C58A	aeternum	
	poG-	A167O	meum	
	posu2	C46O	gloria	
	po-1su2	H269R	alleluia	
	posu1sut1	C80Q	saeculum.	
	torculus			
	to	C78O	Domine	
	tolsc3	C39B	nomine	
	tolsc4lst3	E125P	Domine,	2.166x
	tolst2	C74B	misericordiam	
	to-	G44P	animae	
	to-1	C139A	nostrorum	
	toS	C38A	hodie	
	toS1	E205R	Resurrexi,	0.5x
	toS2	E161S	Sitientes	1.75x
	toS3	C104V	circumfodit	
	toG	G12T	eius	
	toM	C49I	servum	
	toM1	H22S	auxilium	1.71x
	toM3	C39B	mirabile	
	to>	C26A	misericordiam	

Cf.	pe>			
	toS3lst2	C59D	tuae	
	pq!cl-	C87E	sanasti	
	to>1	E130I	obdormiam	2.166x
	to>2	H219T	fortes:	1.63x
	to->	E158D	respexit	2.166x
	toS2sux1	C460	Surge	
	climacus			
	ci	C26A	tuum	
	ci1	G73P	gentes	
	cilsc3	C74B	tuo	
	ci-	C81E	nos	
	ciS	C81E	Domine	
	ciS1	C139A	Deus	
	visut1su2	C53A	Adorabo	
	ciG	C64Q	lapidem	
	ciG1	C42A	ille	
	ciM	C154A	Alleluia	
	ci>	C460	orta	
	cl>	C79L	domum	
	visuv1su2	C35Q	nos	
	visut3	C38T	meo	
	visu1suu1	C26D	laetitia.	
	visut4	C25U	me.	
	cl-1su2	C40A	magna	
	visu1sut2	C43L	Domine	
	visu1suv1	C53S	terrae.	
	visuu1su2	C56D	eius	
	visuu2su2	H103S	salus	
	vi-suu2su2	H103S	o	
	visu2suw1	G113A	Alleluia	0.85x
	visut2su2	G128A	Alleluia	0.85x
	visut3suw1	C153S	dicent.	
	visu1sut1sux1	G126D	salvum	0.85x
	visu1sut1su2	N39H	puer	
	visu1sut1su1sut1	N39H	generavit	
	vi-su1sut1su2	N39H	enim	
	visut1sux1	N59N	filium	
	clSsu1sut1	N65G	tui	
	vi-su4	N68G	Amen.	
	vi-su1sut2su1	N73K	Christe	
	visu1sut1suy1	N221N	Alleluia.	
	visut3suy1	N293A	dicent.	
	visut2suy1	N363C	sacro	1.25x
	visu5	N368L	fornacibus	1.25x
	visu1suu1su2	N372O	erogatione	1.25x
	vi-su1suu2	A174A	meus	
	vi-su1suu1su1	A174A	Alleluia.	
	vi-sut2	H74I	illuminare	
	clS-su2	C46A	eius	
	vi-suu1su2	H75I	sanctus	
	clSsut2	E51R	suscipiant	2.0x
	clSsuu1su2	H390C	surrecturus	1.5x
	vi-sut3	H269D	dicentes	
	vi-su1sut1 *Cf.*	vi-su1sut1su2 minus subpunctis		
	vi-su2sut1	N296V	alleluia	
	vi-su1sux1	E373B	Sanctum	2.133x
	vi-su2sux1	C49I	eum	
	vi-sut4	H16A	servient	1.7x
	vi-suu2	H23E	ecce	1.7x
	vi-suu3	H35G	veniet	1.7x
	visu2sux1	H48B	mater	1.7x
	vi-sut1su2	H22S	elegi	1.71x
	vi-su1suu1	H114C	gloriam	1.71x
	vi-su1suu1su2	H114N	nobis	1.71x
	vi-su1sut2	H324P	aetatis	2.0x
	vi-su3	H307F	natalem	2.0x
	vi-sut1suu1	325I	peregrinatione	2.0x
	scandicus			
	sc	G11P	imperium	0.85x
	sc1 *Cf.*	vipp4 minus first punctum		
	sc-	C123C	terram	
	scS	C154A	Alleluia	
	scG	H179I	Iudas	
	scG1	C52P	Posuisti	
	scM1	C67D	Domine	
	sc>	C26A	Alleluia	
	vi>ppt3	E2R	non	2.166x
	cl>ppt2	H73O	annuntiantes	
	vi-!vi!vi	E155B	Benedicite	2.166x
	vippt3	G140O	ae	0.85x
	vi-pp4	C154A	rectos	
	vi-!vi-!vi	H118G	erubescat	
	vipp4	A127E	eum	
	vi>ppt2	H75V	apertis	
	scG-1	H225M	Mulieres	1.6x
	porrectus flexus			
	pf	C27E	sacrificia	
	pflsc2	C58D	intendentes	
	pf1lsc2	C61I	Ipse	
	pflst2	C122I	fines	
	pf-	C81E	antiquis.	
	pfS	G116C	tribuit	0.85x
	sfM	E248V	alleluia,	2.166x
	pfM	E39E	manere	2.166x
	pf>	C95D	manducare	
Cf.	po>			
	pfMsut1	E34I	eum	2.166x
	pes subbipunctis			
	pesu2	C81E	antiquis.	
	vippt1su2lsc3	C61T	solus:	
	pe-su2	H73S	adoraverunt	
	pe-1su2	C64Q	habitat	
	pe-1sut2	H104T	confitemur	
	pe-1sut1suw1	H219I	videret	1.63x
	peSsu2	A17T	eius	
	peSsut2	C27E	sanctos	
	vippt1su2	C56B	benedicetur.	
	toS2sut1	C139A	Alleluia	
	vi-!ci	E272M	cognovisti	2.166x
	pesu1suw1	G8E	plebem	
	pe-su1sut1	N296V	alleluia	
	pe-su2suy1	N296V	alleluia	
	pe-1sut3	H179A	malum	
Cf.	to>			
	pesu1suu1	C27E	Deus	
	pesu3	C27Q	regis	
	vippt1su1sut1	C40V	omnes	
	vippu1su2	C74B	Altissime.	
	pesu1sut3	C60S	venti.	
	pesu2suv1	C60C	Sana	
	pesu2sut1	E213C	Alleluia.	2.0x
	pesu1sut2	E213C	Alleluia.	2.0x
	vippu1su2lsc3	C32B	caeli	
	pe-1su1suu1	C61I	pascuae	
	peSsu1sux1	G89D	filii	0.85x
	pe-su3sut1	N270A	saecula.	
	pe-1sut1sux1	H72V	orta	
	peSsut3	C460	Jerusalem	
	pesut2	H77S	magi	
	pe-1suw2	E255S	Domini	2.133x
	peSsut1su2	E13P	Domine	2.133x
	pesu1sux1	H25E	angelum	1.7x
	pe-1su1suw1	H29E	deprecabuntur	1.7x
	pesu1sut1 *Cf.*	pesu1sut3 minus 2 tractulis		
	pe-1su1sut1	H306N	de	2.0x
	pe-1su1sut2	H305G	Mariae	2.0x
	pe-1su1suu2	H305G	vidente	2.0x
	pe-1suu1su2	H180U	exhortabamini	1.75x
	pe-su4	C139A	Alleluia	
	pesuv1su1sut1	C139A	Alleluia	
	scandicus flexus			
	sf	C64Q	habitat	
	sflsc3	C27E	Deus	
	sf-	C39B	est	
	sfS	C97D	Deus	
	toppu1	C103C	Cantemus	
Cf.	sfM			
Cf.	sc>			
	to>pp2	E324R	testamentum	2.0x
	torculus resupinus			
	tr	C25U	edoce	
	trlst2	E69P	eius	2.0x

	trM	E336I	resistere	2.166x
	tr>	C82V	nonconfundentur	
	tr~	C60S	Altissimus	
	tr-~	C27Q	et	
	tr-~1	C107A	Alleluia.	
	trS-	C89T	pedes,	
	trS1	E26L	eeiieiue	2.166x
	tr-3	H323D	honore	2.0x
	trsu3	C95D	eius.	
	stropha			
	st	C80Q	commovebitur	
	st-	C52P	Desiderium	
	stS	E263V	est.	2.166x
	st>	C49I	servum	
	distropha			
	ds	C46O	super	
	dslsx6	C45S	forma	
	ds-	C81E	Domine	
	ds>	C38A	Alleluia	
	tristropha			
	ts	C80Q	Dominus	
	tslsx6	C44D	mansuetudinem	
	ts-	C150L	ibimus	
	tsM	C62M	anima	
	ts>	C62M	conculcantes	
	tsM-	A26F	ego	
	trigonus			
	tg	C80Q	aeternum	
	tg1	C81E	antiquis.	
	tgS	C139B	omnibus	
	tgS1	N171D	mei	
	tgS3	C61I	laetitia.	
	tgM	C118C	Dominus	
	tgS-	C118C	Dominus	
	tgS4	C86D	mei	
	bivirga			
	bv	C26D	Propterea	
	bv-	C82B	populus	
	bv>	H33I	montes	
	trivirga			
	tv	C56D	pretioso	
	tv-	C61I	eius.	
	pressus			
	pr	C26A	Alleluia	
	pi	C82R	Domine,	
	prlsc2	E339S	Si	2.166x
	pr-lsc3	C122I	Dei	
	prlst2	C145L	tuo	
	pilsc2	C122I	Dei	
	pilst2	C26A	Alleluia	
	pr-	C59D	Si	
	prM	E173E	insurgentibus	2.166x
	pr>	H72H	in	
	pi>	H103M	Magnus	
Cf.	vs>			
Cf.	or>			
	virga strata			
	vs	C148A	generatio	
	vslsc3	C39B	nomine	
	vslst2	E195D	suis,	2.166x
	vs-	C82B	Domini	
	vs>	C64Q	pennis	
Cf.	ta>			
	vs->	A22V	Rama	
	oriscus			
	or	H112S	candidior	
	or>	C148V	salutari	
	salicus			
	sa	C27E	super	
	sa1	C148A	Alleluia.	
	salsfr1	E3D	Deus	2.166x
	sa-	C25U	tuas	
	sa-1	G30E	tribulationis	
	saG	C107A	Alleluia.	
	saM	H16V	venient	1.7x
	saM1	H23E	homo	1.71x
	saM2	C27Q	regis	
	saM4	E10P	omnis	2.166x
	saM5 *Cf.*	saM5ppt1 minus tractulus		

Cf.	pe>2			
	sa!clS1	C80Q	sicut	
	sa!clS-1	C27Q	tuam	
	sa!clS1lsc2	C46B	Deus	
	sa-lsfr1	C134B	nomen	
	sa1!cl>	G45D	respexit	0.85x
	sa!cl>	G22D	euge	0.85x
	sa!cl	G141O	fortitudo	0.85x
	saM-1	G139A	Domine	0.85x
	sa!cl-	G132V	Alleluia.	0.85x
	saM3	N53A	fratres	
	saM-4	N296V	amoris	
	sa>2	H73S	viderunt	
	sa1!cl->	H32N	eius	1.7x
	sapp1	H320R	venite	2.0x
	saM5ppt1	H392I	de	1.5x
	pes quassus			
	pq	C27E	sacrificia.	
	pqlsfr1	C150L	nostri	
	pq-	C148V	nostro	
	pq>	C84T	pupillo	
	pqS	C52T	maiestate	
	pqS>	H424M	potentiam	
	pq!cl>	G57F	firmamentum	0.85x
	pq!cl	E23T	Domini	2.166x
	quilisma			
	ql	C80Q	commovebitur	
	qi	C103C	Hic	
	ql-	C148A	generatio	
	qi-	H28Q	veniet.	
	ql>	C40V	gentium	
	qi>	C28A	et	
	pes stratus			
	pt	G140O	ae	0.85x

	subpunctis	
	su2 *Cf.*	ci
	su3 *Cf.*	ci1
	su1sut1 *Cf.*	ciS
	sut2 *Cf.*	ciS1
	suu1su2 *Cf.*	visuu1su2
	su1suw1 *Cf.*	ciM
	su1sux1 *Cf.*	ci>
	su1sut2 *Cf.*	visu1sut2
	su1suu1 *Cf.*	visu1suu1
	su2suw1 *Cf.*	visu2suw1
	su2sux1 *Cf.*	vi-su2sux1
	sut1su2 *Cf.*	visut1su2
	sut1suw1 *Cf.*	pe-1sut1suw1
	sut3 *Cf.*	visut3
	suv1su2 *Cf.*	visuv1su2
	sut4 *Cf.*	visut4
	su1suv1 *Cf.*	visu1suv1
	suu2su2 *Cf.*	vi-suu2su2
	suu1su1sut2 *Cf.*	qi-ppt1suu1su1sut2
	sut2su2 *Cf.*	visut2su2
	su1sut3 *Cf.*	pesu1sut3
	su2suv1 *Cf.*	pesu2suv1
	sut3suw1 *Cf.*	visut3suw1
	su1sut1sux1 *Cf.*	visu1sut1sux1
	su1sut1su2 *Cf.*	visu1sut1su2
	su1sut1su1sut1 *Cf.*	visu1sut1su1sut1
	sut1sux1 *Cf.*	visut1sux1
	su4 *Cf.*	vi-su4
	su1sut2su1 *Cf.*	vi-su1sut2su1
	su1sut1suy1 *Cf.*	visu1sut1suy1
	sut3suy1 *Cf.*	visut3suy1
	sut2suy1 *Cf.*	visut2suy1
	su5 *Cf.*	visu5
	su1suu1su2 *Cf.*	visu1suu1su2
	su1suu2 *Cf.*	vi-su1suu2
	su1suu1su1 *Cf.*	vi-su1suu1su1
	su3sut1 *Cf.*	pe-su3sut1
	suu2 *Cf.*	ql-ppt1suu2
	su2sut1 *Cf.*	vi-su2sut1
	su2suy1 *Cf.*	pe-su2suy1
	suw2 *Cf.*	pe-1suw2
	sut2suu1 *Cf.*	ql-ppt1sut2suu1

	su1suv1su2 *Cf.*	ql-ppt2su1suv1su2	
	sut1suu1 *Cf.*	vi-sut1suu1	
	suv1su1sut1 *Cf.*	pesuv1su1sut1	

prepunctis

	pp2 *Cf.*	vi-pp2	
	ppu2 *Cf.*	scS	
	ppt2 *Cf.*	sfS	
	ppu3	C47I	Sede
	ppu4	C58A	Domine
	ppu5	C50G	confregit
	pp4	C78O	escam
	pp3 *Cf.*	pp4	
	ppt3 *Cf.*	vi>ppt3	
	ppt4	N69G	amin.
	ppt5	A56C	videant
	pp1 *Cf.*	vi-pp2	

combined glyphs

	cl!pi	C82R	Domine	
	cl!pilsc1lst3	C89T	manum	
	cl!pilsc2	C149A	cantabo	
	cl!pr	G51S	ipsa	0.85x
	cl-!pi	E35E	illis	2.166x
	ql!cl	C148A	Domine	
	ql!cl!cl> *Cf.*	ql!cl!cl>ppt1 minus tractulus		
	ql!cl!pi *Cf.*	ql!cl!pippt1 minus tractulus		
	ql!pf-	G42D	mane	0.85x
	ql!pf	G35D	Domine	
	ql!cl-	C26A	Alleluia	
	ql!clM *Cf.*	ql!clMppt1 minus tractulus		
	ql!cl-> *Cf.*	ql!cl->ppt1 minus tractulus		
	ql!cl>	H26L	salva	0.5x
	ql!po	C148V	nostro	
	ql!po-1	C148A	Alleluia.	
	ql!po>	H33I	terram	
	ql!vs	C148A	refugium	
	ql!vssu2 *Cf.*	ql!vsppt1su2 minus tractulus		
	ql!vssut2 *Cf.*	ql!vsppt1sut2 minus tractulus		
	qi!vs *Cf.*	qi!vssu2 minus subbipunctis		
	qi!vssu2 *Cf.*	qi!vsppt1su2 minus tractulus		
	qi!vssu1sut2 *Cf.*	qi!vsppt1su1sut2 minus tractulus		
	ql-su1suu1 *Cf.*	ql-ppt1su1suu1 minus tractulus		
	ql-su1suv1su2 *Cf.*	ql-ppt2su1suv1su2 minus tractulis		
	qisu2	E166C	sermones	2.0x
	qisut2 *Cf.*	qippt2sut2 minus tractulis		
	ql-su3 *Cf.*	ql-ppt1su3 minus tractulus		
	qlsut3 *Cf.*	qlppt1sut3 minus tractulus		
	qlsu2 *Cf.*	qlppt1su2 minus tractulus		
	ql-su2 *Cf.*	ql-ppt1su2 minus tractulus		
	ql-sut2	H73S	cedebat	
	ql-su1sux1 *Cf.*	ql-ppt1su1sux1 minus tractulus		
	ql-su1sut2 *Cf.*	ql-ppt1su1sut2 minus tractulus		
	ql-sut1sux1	H72H	complacui	
	ql-suu2 *Cf.*	ql-ppt1suu2 minus tractulus		
	ql-sut2suu1 *Cf.*	ql-ppt1sut2suu1 minus tractulus		
	qi-su2	H73S	cedebat	
	qi-su1sut2 *Cf.*	qi-ppt1su1sut2 minus tractulus		
	qi-suu1su1sut2 *Cf.*	qi-ppt1suu1su1sut2 minus tractulus		
	qi-su1sux1	H26G	certamine	1.7x
	ql-su1suu2	H34F	facinora	1.71x
	qi!cl	C82B	spiritu	
	qi!cl-	C148A	factus	
	qi!cl>	H77S	Magi	
	qi!po	H87E	Deus	
	qi!po>	H173C	adversum	1.9x
	pf!clM-ppt1	E327O	narante	2.166x
	cl!po>ppt1	E327O	narante	2.166x
	qlppt1	C38T	Sede	
	qippt1	C37A	Domine	
	qi!clppt1	C88T	hostias,	
	ql-ppt1	C26D	laetitiae.	
	ql>ppt1	C38T	virtutis	
	qi>ppt1	C60S	terram.	
	qisut1sux1	H269S	linguarum	
	qlsut2	H269D	alleluia	
	qi-sut2	H269D	alleluia	
	qi-ppt1su1sut2	H269R	magnalia	

	ql!cl!cl>ppt1	E133I	corda	2.166x
	ql!cl!pippt1	E31T	maris	2.166x
	ql!cl-ppt1	C40A	terram.	
	ql!clMppt1	E324R	tuum	2.166x
	ql!cl->ppt1	G132A	Alleluia.	0.85x
	ql!clS-2ppt1sut1	E3D	et.	2.0x
	ql!clS-2sut1 *Cf.*	ql!clS-2ppt1sut1 minus tractulus		
	ql!clS-2ppt1 *Cf.*	ql!clS-2ppt1sut1 minus tractulus		
	ql!clS-2 *Cf.*	ql!clS-2ppt1sut1 minus 2 tractulis		
	ql!poppt1	C59D	Domine.	
	ql!po-1ppt1	C58A	homo.	
	ql!po~ppt1	H179L	pauperem	
	ql!po~ *Cf.*	ql!po~ minus tractulus		
	ql-ppt1su2	H72V	Jerusalem	
	ql-ppt1su1sux1	H73O	Reges	
	ql-ppt1su1sut2	H75I	patris	
	ql-ppt1suu2	H72H	Hic	
	ql-ppt1sut2suu1	H16M	altissimi	1.7x
	ql!vsppt1su2	H73O	Domino	
	ql!vsppt1 *Cf.*	ql!vsppt1su2 minus subbipunctis		
	ql!vsppt1sut2	H73S	gaudio	
	qi!vsppt1su2	H74I	Domini	
	qi!vsppt1su1sut2	H392L	viderent	1.5x
	qi-sut3	H322B	in	2.0x
	ql-ppt1su1suu1	H305H	David	2.0x
	ql-ppt1su3	H323D	subire	2.0x
	qlppt1sut3	H305G	Redemptorem	2.0x
	qlppt1su2	E166C	secundum	2.0x
	ql-ppt2su1suv1su2	H306N	illustrat	2.0x
	ql-su1sut1	H307F	iustitiae	2.0x
	qippt2sut2	H297B	generationes	2.0x
	ql!vssu3	H443V	ei	2.0x
	to!ciG	E51R	eum	2.0x
	to!ciG	C28T	gloriae.	
	to!cl	C84T	pupillo	
	to!cl!cl!cl	C64Q	sepavit,	
	to!cl!po	E31T	veritas	2.166x
	to!cl!po-1	N100T	veritas	
	to!cl!pi	C45S	mea	
	to!cl-	N86D	propterea	
	to!cl-!cl-	C54A	Adducentur	
	to!cl-1	C45S	filiis	
	to!clS	C43A	terram.	
	to!cl!cl	C43A	Domini,	
	to!po	C64Q	sagitta	
	to!posu2	C49I	eo,	
	to!pr	G11V	Deo	0.85x
	to!pi	C60C	tui.	
	toSsu2	H72H	mihi	
	pf!cl-	G51I	meus	
	pf!clM- *Cf.*	pf!clM-ppt1 minus tractulus		
	pf!pr	G9A	Alleluia	
	cl!po> *Cf.*	cl!po>ppt1 minus tractulus		
	cl!cl>1	N67G	mundi	
	cl!clSsut2	E256C	psallite	2.166x
	cl!clS *Cf.*	cl!clSsut2 minus 2 tractulis		
	cl!po	C26A	nobis.	
	cl!po-1	H119C	in	
	tr-1su2	H73O	dona	
	trSsut2	H305H	devotissime	2.0x
	ci-pp2	H73S	magno	
	toS2su2	H73S	eam	
	toS2sut2	C150L	dicta	
	to-1su2	H322C	ostendere	2.0x
	to-1su1	H323B	Gallus	2.0x
	to-1suu1	H173I	me	1.8x
	to-1suu1su2	H324V	cognovi	2.0x
	po-1pp2su1sux1	H323D	desiderium	2.0x
	po-1su1sux1 *Cf.*	po-1pp2su1sux1 minus bipunctis		
	pisu1	H307N	annuntiavit	2.0x
	pisut1	C153B	Alleluia	
	pq-su2	C132D	praetioso	
	pe!vi	C57B	qui	
	tgsuu1	C49I	eum.	
	tgSsuu1	C57B	nimis.	
	bv-sut2	H179U	fuisset.	1.75x
	sa-su2sut1	C49I	eum	

	significative letters				
	lsam	C87E	abstraxisti	altius	
	lsal	E20E	Sion,	altius mediocriter	2.166x
	lsb	C39B	Domino	bene	
	lsc	C27E	eius	celeriter	
	lscw	C148V	nostro.	celeriter (wide form)	
	lsco	C134B	nomen	coniunguntur	
	lscm	C58A	oblivio	celeriter mediocriter	
	lsd	E11C	Virgo	deprimatur	1.85x
	lse	C84T	adiutor	equaliter	
	lsew	H266D	si	equaliter (wide form)	
	lseq	C64Q	viis	equaliter	
	lsfid	E16B	Domino	fideliter	2.166x
	lsfr	C58A	non	frendor	
	lsg	C27Q	ovem	gutture	
	lsi	C87E	Domine	iusum	
	lsim	E336I	universa	iusum mediocriter	2.166x
	lsiv	E92I	misericors	iusum valde	2.166x
	lsk	C62M	confidit	klenge	
	lsl	C38T	Tecum	levare	
	lslb	C58A	non	levare bene	
	lslc	E151A	ab	levare celeriter	2.166x
	lslen	C59D	servitui.	leniter	
	lslm	C38T	Tecum	levare mediocriter	
	lslp	E161S	dicit	levare parvum	2.166x
	lslt	E148L	Laudate	levare tenete	2.0x
	lsm	C38T	principium	mediocriter	
	lsmoll	E33V	fines	molliter	2.166x
	lsp	C64Q	altissimi	parvum	
	lspar	E104S	miserere	paratim	2.166x
	lspfec	E2D	fructum	perfecte	2.166x
	lspm	C40V	suam.	parvum mediocriter	
	lspulcre	E64L	hac	pulcre	2.0x
	lss	C27E	sanctos	sursum	
	lssb	C40V	iustitiam	sursum bene	
	lssc	C84T	Tibi	sursum celeriter	
	lssimil	E193D	ne	similiter	2.166x
	lssimul	E42A	sumus.	simul	2.166x
	lssm	C59D	Si	sursum mediocriter	
	lsst	C87E	Domine	sursum tenere	
	lssta	C27E	Congregate	statim	
	lst	C39B	Domino	tenete	
	lstw	C58A	non	tenete (wide form)	
	lstb	C39B	Domino	tenete bene	
	lstm	E3D	tu	tenete mediocriter	2.166x
	lsv	E131O	semper	valde	2.0x
	lsvol	C58A	aeternum	volubiliter	
	lsx	C87E	me:	expectare	

manuscript sources of the grelaon font

The list of glyphs uses following abbreviations for manuscript references; the page number is always for the start of some antiphon, responsory, mass propers, even when the actual neume is on a later page. At the end of lines, GT stands for Graduale Triplex, Solesmes, 1979, OT for Offertoriale Triplex, Solesmes, 1985, GN for Graduale Novum de Dominicis et Festis, Conbrio Verlag Regensburg, 2011, followed by page number which also contains the corresponding propers, antiphons or responsories.

All the grelaon font glyphs taken from the LAON_BM_ms239 codex with the permission from the Bibliothèque municipale de la ville de Laon http://manuscrit.ville-laon.fr, the picture fragments from there are only present in the font source file to help drawing the glyphs.

Graduale Laudunensis, codex 239

L1 = http://manuscrit.ville-laon.fr/_app/_visualisation.php?cote=Ms239&vue=9

L8U Universi qui te GT16
L8A Ad te Domine levavi OT5
L8D Dominus dabit GT17
L9E Ex Sion species GT18
L10I Ierusalem surge GT20
L11Q Qui sedes, Domine GT22
L13A Ave Maria GT36
L14P Prope es tu Domine GT24
L14O Ostende nobis GT31
L17H Hodie scietis GT38
L18L Laetentur caeli GT44
L19B Benedictus qui venit GT45
L20P Puer natus est GT47
L24D Dextera Domini GT194
L24T Timebunt gentes GT265
L25L Laetabitur iustus GT473
L25M Mirabantus omnes GN232
L28G Gaudeamus omnes GT545
L31D De profundis GT673
L33S Sciant gentes GT88
L33C Commovisti Domine GT89
L33P Perfice gressus OT90
L34E Esto mihi in Deum GT275
L35T Tu es Deus GT244
L37M Misereris omnium GT62
L37S Miserere mei Deus GT63
L38I Iacta cogitatum GT285
L40Q Qui habitat GT73
L43L Levabo oculos meos GT78
L43S Scapulis suis GT77
L43I Sicut oculi servorum GT77
L44D Domine refugium GT79
L44I Dirigatur oratio GT340
L45R Reminiscere miserationum GT81
L53C Confitemini Domino GT158
L54A Adiutor meus GT86
L54B Benedicam Dominum OT88
L56D Deus in adiutorium GT315
L56S Salvum fac populum GT354
L56I Iustus Dominus GT93
L56P Propitius esto Domine GT288
L58A Ad Dominum dum tribularer GT282
L60E Exsurge Domine GT96
L60A Ad te levavi GT97
L61I Iustitiae Domini GT309
L64O Oculi omnium GT343
L64D Domine fac mecum OT37
L67Q Qui biberit aquam GT99
L67S Si ambulem GT125
L68N Nemo te condemnavit GT124
L68Q Qui confidunt GT109
L71D Dum santificatus fuero GT249
L72B Benedicite gentes GT231
L75V Videns Dominus GT124
L76E Eripe me Domine GT121
L77S Saepe expugnaverunt GT122
L77C Confitebor tibi Domine GT123
L78H Hoc corpus GT170
L80S Sperent in te omnes GT286
L81E Exaltabo te Domine GT112
L81R Eripe me de inimicis GT129
L82O Omnia quae fecisti GT342
L82T Tollite hostias GT272
L84B Benedictus es, Domine OT48
L90D Deus Deus meus GT144
L92E Exsurge Domine GT150
L93E Erubescant et revereantur GT152
L93G Ego autem dum mihi GT152
L94C Custodi me Domine GT154
L95D Domine, exaudi orationem GT172
L99P Populе meus quid GT176
L101V Vinea facta est GT188
L103R Resurrexi GT196
L103H Haec dies GT196
L103P Alleluia Pascha nostrum GT197
L103T Terra tremuit GT199
L104I Introduxit vos Dominus GT200
L105A Angelus Dominus GT217
L105N Angelus Dominus GT201
L106S Surrexit Dominus GT207
L107H Haec dies GT206
L109P Populus acquisitionis GT210
L112D Deus Deus meus GT224
L112O Omnes qui in Christo GT61
L113E Ego sum pastor GT224
L114C Cantate Domino GT225
L114V Vocem iucunditatis GT229
L117R Repleti sumus GT441
L118E Ego sum vitis vera GT228
L122N Non vos relinquam GT232
L122V Viri Galilaei OT172
L123A Ascendit Deus GT237
L123E Exaudi Domine vocem GT241
L125U Ultimo festivitatis GT251
L125S Spiritus Domini GT252
L126F Factus est repente GT256
L128S Spiritus ubi vult spirat GT218
L132E Ecce quam bonum GT351
L136A Amen dico vobis quod vos GT436
L138S Semel iuravi GT492
L144C Clamaverunt iusti GT454
L149E Ego dixi Domine GT279
L149C Cantabo Domino qui bona GT283
L152S Sicut in holocausto GT299
L152I Inclina aurem tuam GT300
L164D De profundis GT368
L168I Iubilate Deo GT258
L168S Specie tua GT416
L168L Laudate Deum GT262
L169A Adorabo ad templum GT270
L173D Deus iudes iustus GT286
L174D Domine in virtute tua GT292
L174I In te Domine speravi GT296
L174O Omnes gentes plaudite manibus GT298
L174T Te dece hymnus GT305

	uncinus		
	un	L78H	testamenti
	un1	L78H	calix
	un2 *Cf.*	un2lsm1	minus letter
	un2lsm1	L82O	fecisti
	unlsm1	L75V	qui
	unlst1	L103R	alleluia
	unlst2	L82O	fac
	unlst8	L24D	opera
	un1lst8	L112O	Christo
	unlsm2lss3lst8	L43S	scuto
	punctum		
	pu	L82O	iudicio
	pu1	L56D	revereantur
	pulsnt3	L75V	flentes
	virga		
	vi	L82O	Domine
	viS *Cf.*	vipp2su2	minus punctis
	vi>	L78H	in
	vi>1 *Cf.*	vi>1lsa2lsc8	minus letters
	vi>2	L82O	nobiscum
	vilst1	L78H	meam
	vilst2	L95D	meus
	vilst9	L105A	quaeritis,
	vilss1	L103P	Alleluia.
	vilss2	L95D	avertas
	vilsst1	L103P	Alleluia.
	vilst2lsa8	L95D	eius.
	vi>lsa6	L78H	testamenti
	vi>lsc6	L56D	adiuvandum
	vi>1lsa2lsc8	L126F	loquentes
	tractulus		
	ta	L24T	tuam.
	ta>	L78H	dicit
	ta>1	L82O	mandatis
	ta>2	L90D	sancto
	ta>lsnl2	L45R	dominentur
	ta>2lsc2	L90D	Libera
	pes		
	pe	L78H	commemorationem
	peG	L78H	vobis
	pe>	L95D	exaudi
	pe>1	L78H	sanguine
	peG1	L67S	tua,
	pelsa2	L82O	peccavimus
	pelsc2	L67S	tuus,
	pelsl2	L43L	iustitiam
	pelsm2	L81E	meam,
	pelsn2	L149C	qui
	pelsn3	L125S	replevit
	pelsst1	L103P	immolatum
	pelsst1lsm4	L44I	Elevatio
	pelst2	L103H	saeculum
	peGlsf1	L67S	non
	peGlsa6	L109P	vocavit
	pe>1lsa6	L114V	alleluia
	porrectus		
	po	L78H	facite
	poG	L64D	mecum
	poM *Cf.*	poMsun1	minus uncinus
	po> *Cf.*	po>pp2	minus prepunctis
	po>1	L81E	descendentibus
	polsf2lst3	L95D	meus
	polst2	L103H	saeculum
	clivis		
	cl *Cf.*	cllsn9	minus letter
	clG	L78H	hoc
	clG1	L105N	accedens
	clGlsa5	L82O	obedivimus
	clGlsnl2	L44I	Elevatio
	clM	L78H	quotiescumque
	cl>	L75V	monumentum
	cl>1	L34E	locum
	cllshn8	L64O	imples
	cllsn8	L95D	velociter
	cllsn9	L67S	tua,
	cllsst8	L64O	imples
	cllst9	L95D	tempus

	cl>lsa5	L71D	effundam
	oriscus-clivis		
	oc	L78H	sanguine
	ocG	L56D	Deus
	ocG1	L11Q	veniet
	ocGlsa5	L103T	resurgeret
	oclst8	L95D	Domine
	torculus		
	to	L78H	pro
	toS	L56S	populum
	toS1	L68N	Nemo
	toG	L78H	tradetur
	to>	L67S	mortis
	to	L37M	et
	tolsf4	L95D	tempus
	tolsn8	L44I	vespertinum.
	tolst9	L103P	Alleluia.
	tolsth8	L56I	Dominus,
	tolsnt2lst9	L92E	me
	toGlsa1lst8	L109P	alleluia.
	toGlsa5	L68N	condemnabo
	toSlsa5	L77S	enim
	to>lsa5	L118E	in
	to>2lsa3	L67Q	salientis
	to>2lsc1lsa3	L104I	terram
	climacus		
	ci	L95D	eius.
	unsun2	L82O	da
	unsun3	L103P	immolatus
	unsux1	L169A	sanctum
	unsun1su2	L109P	tenebris
	pusu1sun1	L78H	tradetur
	pusu1sun2	L95D	Sion
	pusu2sun1	L95D	eius.
	visu1sun1	L95D	faciem
	visu1sun2	L40Q	latere
	visu1sun2lsa5	L174D	et
	visu1sun2lsa6	L174D	et
	visu1sun1su1 *Cf.*	visu1sun1su1lst6 minus letter	
	visu1sun1su1lst6	L105N	descendit
	visu1sun1su2 *Cf.*	visu1sun1su2lss1lst6 minus letters	
	visu1sun1su2lss1lst6	L168I	Alleluia.
	visu2	L95D	misereberis
	visun1su1sun1	L60E	conspectu
	visun1su1sun1lst6	L35T	virtutem
	cilsf1	L95D	Sion
	cilsn8	L58A	iniquis,
	pusu1sun1lsf1	L95D	mei
	pusu1sun2lsa5	L67S	timebo
	pusu1sun1lst7	L60E	homo
	pusu3	L64O	escam
	pusu4	L92E	persequntur.
	unsun2su1	L64O	escam
	unsun1su1 *Cf.*	unsun1su1lst2lsc6 minus letters	
	unsun1su1lst2lsc6	L11Q	deducis
	unsun2lsa5	L19B	mirabile
	unsun3lsa3	L24T	gentes
	unsun3lsa6	L24T	tuam.
	unsun2su2	L169A	confitebor
	un1sun2su2	L136A	aeternam
	un1sun1su1sun1lsa1lsc4	L60E	retrorsum
	un1sun1su1sun1lsa3lsc6	L60E	retrorsum
	un1sun1su2 *Cf.*	un1sun1su2lsa5 minus letter	
	un1sun1su2lsa5	L107H	Domini
	scandicus		
	sc *Cf.*	sclst2 minus letter	
	sc1	L82O	quia
	sclst2	L78H	quotiescumque
	sc1lsc4	L109P	vocavit
	porrectus flexus		
	pf	L78H	corpus
	pf1	L95D	exaudi
	pf2	L101V	Sorec,
	pfS	L56D	festina
	pfM *Cf.*	pfMlsmd3 minus letter	
	pfMlsmd3	L81E	meam,
	pf>	L164D	ad
	pf>1	L90D	miserunt
	pf>2	L104I	alleluia,
	pf	L95D	clamor
	pf 1	L31D	ad
	pes subbipunctis		
	tosu1	L82O	Domine
	tosun1	L95D	fumus
	tosun2	L105N	revolvit
	vippn1su2	L82O	misericordiae
	topp1su1 *Cf.*	topp1su1lsa6 minus letter	
	topp1su1lsa6	L95D	ad
	pesu2 *Cf.*	pesu2lst3 minus letter	
	pesu2lst3	L152I	eruas
	pesun3 *Cf.*	pesun3lss1lsa3 minus letters	
	pesun3lss1lsa3	L67S	medio
	pesu2sun1	L24T	terrae
	tosun1lsa5	L67S	tua,
	tosu1lsc5	L19B	venit
	toSsun1lsa5	L138S	luna
	tosun1su1sun1	L33S	venti.
	tosun1su2lsn3lst6	L112D	nomine
	tosun2lsa6	L123A	alleluia.
	scandicus flexus		
	sf	L78H	meo
	sfG	L33C	eam.
	torculus resupinus		
	tr	L78H	sumitis
	trS	L95D	Domine,
	tr>	L64D	diligerent
	poppn1	L82O	vero
	po>ppn1	L82O	secundum
	po>ppn1ppz1	L18L	faciem
	po>pp2	L28G	angeli,
	trlst3lshp7	L109P	alleluia.
	trlss1lss3	L103P	Christus.
	trSlsc1	L103P	nostrum
	distropha		
	ds *Cf.*	dslst2 minus letter	
	dslst2	L95D	meam
	dslst8	L71D	effundam
	ds1 *Cf.*	ds1pp3 minus prepunctis	

tristropha

	ts	L82O	fac
	ts1	L76E	Eripe
	ts>	L56D	adiuvandum
	ts>1	L103R	alleluia
	tslsa2	L13A	tecum
	tslsc2lst2	L58A	Domine,
	ts>lsc2	L103R	alleluia
	ts>lsc6	L103T	dum

trigonus

	tg	L67S	tua,
	tglsn8	L67S	mala
	tglst9	L103R	alleluia

bivirga

	bv	L82O	Domine
	bvlsa2	L82O	nobis

trivirga

	tv	L103H	bonus
	tvlsm2	L44I	Elevatio

pressus

	pr	L95D	veniat
	pi *Cf.*	cl1!pi	minus clivis

virga strata

	vs	L64D	Deus
	vsG	L64D	suavis

oriscus

	or *Cf.*	ocG	minus uncinus
	orlst3	L103P	Alleluia.
	or1 *Cf.*	ocG1	minus uncinus
	or> *Cf.*	or>lsa1	minus letter
	or>lsa1	L103R	alleluia,

salicus

	sa	L82O	nomini
	sa1	L78H	novi
	sa>1	L9E	sanctos
	salsc4	L93G	exsurge
	salsc6	L93G	autem,
	sa1lsc4	L81E	salvasti
	sa1lsc6	L113E	bonus,

pes quassus

	pq	L103H	eius.
	pq1	L169A	Alleluia. L174T

quilisma

	ql *Cf.*	qllsst2	minus letter
	qllsst2	L53C	Domine,
	ql> *Cf.*	ql>ppn1	minus uncinus

pes stratus

	pt1	L13A	ventris
	pt2	L8U	confundentur,
	pt3	L94C	Domine.

subpunctis

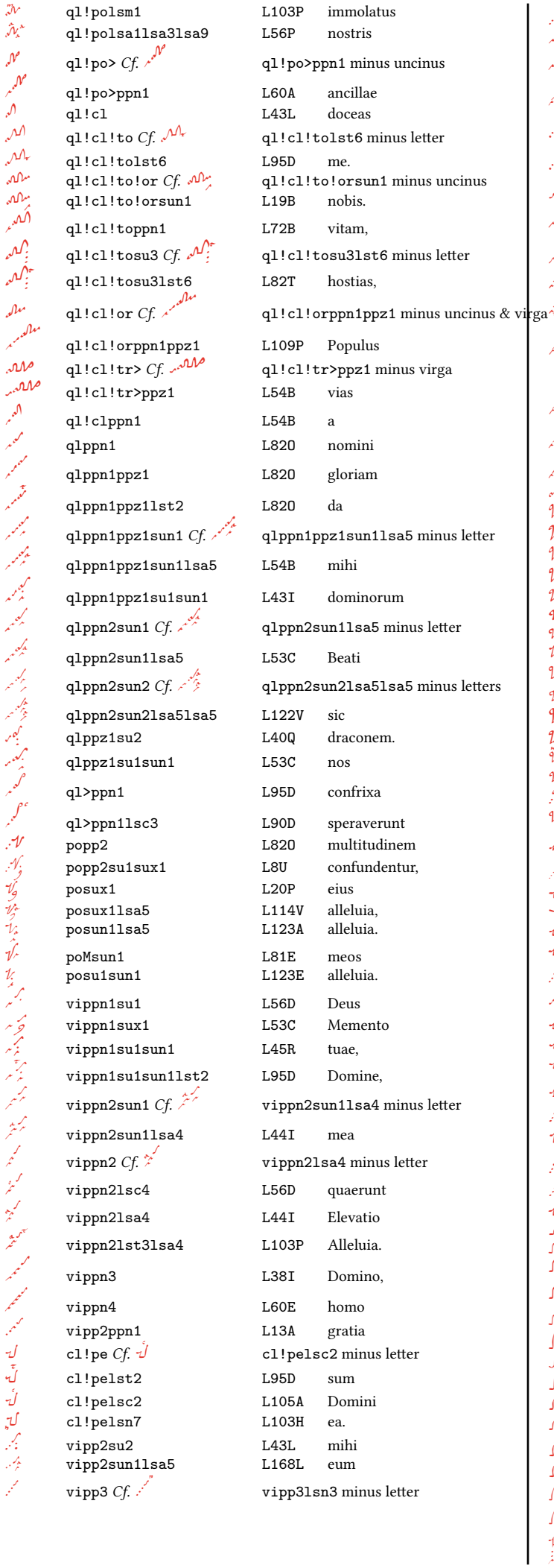
	su1 *Cf.*	cl1!pi!tosu1
	su1sun1 *Cf.*	cl!toppz1su1sun1
	su1sun1su1 *Cf.*	visu1sun1su1lst6
	su1sun1su2 *Cf.*	visu1sun1su2lss1lst6
	su1sun2 *Cf.*	pusu1sun2
	su1sux1 *Cf.*	popp2su1sux1
	su2 *Cf.*	cl1!pi!pesu2
	su2sun1 *Cf.*	clsu2sun1lst6
	su3 *Cf.*	pusu3
	su4 *Cf.*	pusu4
	sun1 *Cf.*	cl1!pi!orsun1
	sun1su1 *Cf.*	unsun1su1lst2lsc6
	sun1su1sun1 *Cf.*	visun1su1sun1
	sun1su2 *Cf.*	qlppn1sun1su2
	sun2 *Cf.*	qlsun2
	sun2su1 *Cf.*	unsun2su1
	sun2su2 *Cf.*	unsun2su2
	sun3 *Cf.*	unsun3
	sun4 *Cf.*	bvsun4
	sux1 *Cf.*	posux1

prepunctis

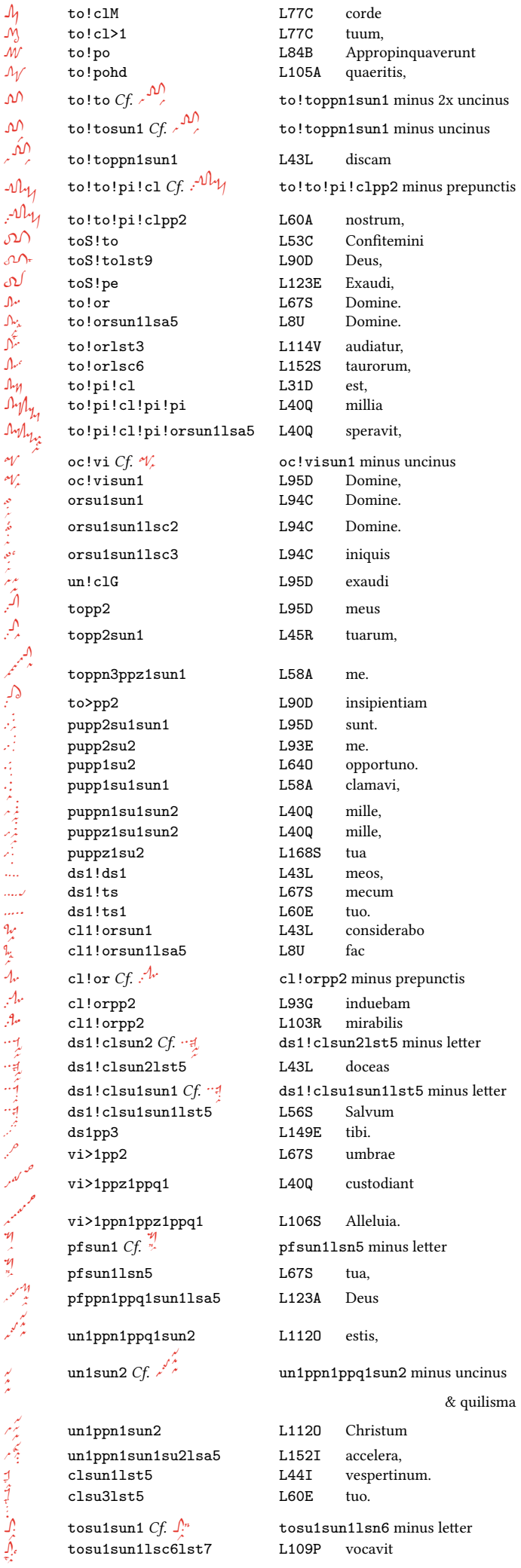
	pp1 *Cf.*	pupp1su2
	pp2 *Cf.*	cl1!pipp2
	pp2ppn1 *Cf.*	vipp2ppn1
	pp3 *Cf.*	vipp3lsn3
	pp4 *Cf.*	vipp4lst2
	ppn1 *Cf.*	po>ppn1
	ppn1pp1 *Cf.*	vippn1pp1
	ppn1pp1ppz1 *Cf.*	vippn1pp1ppz1lst8
	ppn1ppq1 *Cf.*	vippn1ppq1
	ppn1ppz1 *Cf.*	ql!cl!orppn1ppz1
	ppn1ppz1ppq1 *Cf.*	vi>1ppn1ppz1ppq1
	ppn2 *Cf.*	vippn2lst3lsa4
	ppn2ppz1 *Cf.*	vippn2ppz1sun1lsa5
	ppn2ppz1ppq1 *Cf.*	vippn2ppz1ppq1lst2lsa8
	ppn3 *Cf.*	unppn3sun1lsa5
	ppn3ppz1 *Cf.*	toppn3ppz1sun1
	ppn4 *Cf.*	vippn4
	ppq1 *Cf.*	vippq1
	ppz1 *Cf.*	puppz1su2
	ppz1ppq1 *Cf.*	vi>1ppz1ppq1

combined glyphs

	pe!ta *Cf.*	pe!talsa2	minus letter
	pe!talsa2	L82O	iudicio
	pe!talsc1lst2	L114C	revelavit
	pe!un *Cf.*	pe!unlsc2	minus letter
	pe!unlsc2	L95D	meam
	pe!pu	L95D	eius.
	pe!ta>2 *Cf.*	pe!ta>2lsa2	minus letter
	pe!ta>2lsa2	L60E	gentes
	pe!or	L93G	induebam
	qlppn1sun1	L53C	Domino,
	qlppn1sun1su2	L82O	tibi
	qlppn1su1sun2	L95D	frixorio
	qlppn1su2 *Cf.*	qlppn1su2lsnl1	minus letters
	qlppn1su2lsnl1	L75V	mortuus.
	qlppn1sun1lsc5	L54B	quoniam
	qlppn1su1sun1	L107H	fecit
	qlsu1sun1	L43L	tua
	qlsu1sux1	L71D	aquam
	qlsu2	L67S	tua,
	qlsux1	L53C	quoniam
	qlsun1su2	L44I	mearum
	qlsun1 *Cf.*	qlsun1lsa5	minus letter
	qlsun1lsa5	L40Q	millia
	qlsun2	L122N	alleluia.
	qlsun2lsa5	L125U	alleluia.
	ql!po	L82O	obedivimus
	ql!poppz1 *Cf.*	ql!poppz1lsa3	minus letter
	ql!poppz1lsa3	L95D	Domine,
	ql!poppn1	L43L	Domine,
	ql!posu1sun1 *Cf.*	ql!poppz1su1sun1	minus uncinus
	ql!poppz1su1sun1	L43L	mandata
	ql!posun1 *Cf.*	ql!poppn1sun1	minus uncinus
	ql!poppn1sun1	L61I	Iustitiae
	ql!poppn1lst2	L103P	Alleluia.
	ql!posux1 *Cf.*	ql!poppn1sux1	minus uncinus
	ql!poppn1sux1	L61I	corda,



ql!polsm1	L103P	immolatus
ql!polsa1lsa3lsa9	L56P	nostris
ql!po> *Cf.*	ql!po>ppn1 minus uncinus	
ql!po>ppn1	L60A	ancillae
ql!cl	L43L	doceas
ql!cl!to *Cf.*	ql!cl!tolst6 minus letter	
ql!cl!tolst6	L95D	me.
ql!cl!to!or *Cf.*	ql!cl!to!orsun1 minus uncinus	
ql!cl!to!orsun1	L19B	nobis.
ql!cl!toppn1	L72B	vitam,
ql!cl!tosu3 *Cf.*	ql!cl!tosu3lst6 minus letter	
ql!cl!tosu3lst6	L82T	hostias,
ql!cl!or *Cf.*	ql!cl!orppn1ppz1 minus uncinus & virga	
ql!cl!orppn1ppz1	L109P	Populus
ql!cl!tr> *Cf.*	ql!cl!tr>ppz1 minus virga	
ql!cl!tr>ppz1	L54B	vias
ql!clppn1	L54B	a
qlppn1	L82O	nomini
qlppn1ppz1	L82O	gloriam
qlppn1ppz1lst2	L82O	da
qlppn1ppz1sun1 *Cf.*	qlppn1ppz1sun1lsa5 minus letter	
qlppn1ppz1sun1lsa5	L54B	mihi
qlppn1ppz1su1sun1	L43I	dominorum
qlppn2sun1 *Cf.*	qlppn2sun1lsa5 minus letter	
qlppn2sun1lsa5	L53C	Beati
qlppn2sun2 *Cf.*	qlppn2sun2lsa5lsa5 minus letters	
qlppn2sun2lsa5lsa5	L122V	sic
qlppz1su2	L40Q	draconem.
qlppz1su1sun1	L53C	nos
ql>ppn1	L95D	confrixa
ql>ppn1lsc3	L90D	speraverunt
popp2	L82O	multitudinem
popp2su1sux1	L8U	confundentur,
posux1	L20P	eius
posux1lsa5	L114V	alleluia,
posun1lsa5	L123A	alleluia.
poMsun1	L81E	meos
posu1sun1	L123E	alleluia.
vippn1su1	L56D	Deus
vippn1sux1	L53C	Memento
vippn1su1sun1	L45R	tuae,
vippn1su1sun1lst2	L95D	Domine,
vippn2sun1 *Cf.*	vippn2sun1lsa4 minus letter	
vippn2sun1lsa4	L44I	mea
vippn2 *Cf.*	vippn2lsa4 minus letter	
vippn2lsc4	L56D	quaerunt
vippn2lsa4	L44I	Elevatio
vippn2lst3lsa4	L103P	Alleluia.
vippn3	L38I	Domino,
vippn4	L60E	homo
vipp2ppn1	L13A	gratia
cl!pe *Cf.*	cl!pelsc2 minus letter	
cl!pelst2	L95D	sum
cl!pelsc2	L105A	Domini
cl!pelsn7	L103H	ea.
vipp2su2	L43L	mihi
vipp2sun1lsa5	L168L	eum
vipp3 *Cf.*	vipp3lsn3 minus letter	
vipp3lsn3	L152S	Sicut
vippn1pp1	L11Q	Israel,
vippn1pp1su1sun2	L40Q	volante
vippn3su4	L173D	Alleluia.
vipp4 *Cf.*	vipp4lst2 minus letter	
vipp4lst2	L13A	Ave
vippz1ppq1	L40Q	lapidem
vippn1ppq1	L33S	venti.
vippn1ppz1ppq1	L101V	Israel
vippn1pp1ppz1 *Cf.*	vippn1pp1ppz1lst8 minus letter	
vippn1pp1ppz1lst8	L25M	ore
vippn2ppz1ppq1 *Cf.*	vippn2ppz1ppq1lst2lsa8 minus letters	
vippn2ppz1ppq1lst2lsa8	L169A	confitebor
vippn2ppz1sun1lsa5	L173D	Alleluia.
vippn3sun1lsa5	L173D	Alleluia.
vippq1	L168S	regna.
cl1!or *Cf.*	cl1!orlss3 minus letter	
cl1!orlss3	L60E	iudicentur
cl1!orlsc3	L60E	conspectu
cl1!pi	L95D	avertas
cl1!pi>	L53C	misericordia
cl1!pi!to	L82O	fecisti
cl1!pi!vi	L95D	Domine,
cl!pi!vi>	L67S	Virga
cl1!pi!cl	L64O	implebis
cl1!pi!pesu2	L61I	dulciora
cl1!pilsm3lsa6	L174I	me
cl1!pilsn8	L53C	iudicium,
cl1!pilsn9	L64O	Aperis
cl1!pilst9	L90D	Longe
cl1!pipp2	L60A	Deum
cl1!pi!orsun1	L90D	Israel.
cl!po	L67S	es,
cl!popp3	L122V	ascendentem
cl!to	L17H	eius.
cl!tosu1	L17H	eius.
cl!tosu1lsc6	L81E	me
cl!tosu1lsc6lst8	L81E	me
cl!topp2	L64D	misericordia
cl!toppz1su1sun1	L35T	Israel
cl!tolst9	L56I	dilexit
cl!to>	L56I	aequitatem
cl!tosun1lsa5	L164D	meam.
cl!to!pi *Cf.*	cl!to!pipp2 minus prepunctis	
cl!to!pipp2	L33C	arcus,
cl!to!pi!cl	L77S	Israel
cl!to!pi!clpp2	L33C	eius,
cl!to!pi!clpp2lsn9	L68Q	Sion
cl!to!cl!cl	L81R	meis,
to!cl	L95D	Domine,
to!cl!cl *Cf.*	to!cl!cllst9 minus letter	
to!cl!clsun2 *Cf.*	to!cl!clsun2lsa5 minus letter	
to!cl!clsun2lsa5	L43L	tuam
to!cl!cllst9	L8A	invocavite.
to!cl!clsu1	L33P	meam.
to!cl!pi	L84B	iniqui
to!cl!orsun1	L114V	terrae
to!clsun1	L54B	quoniam
to!clsun1lsa5	L11Q	Ioseph.
to!clsu2 *Cf.*	to!clsu2lsn6 minus letter	
to!clsu2lsn6	L8D	dabit
to!clsu1sun1 *Cf.*	to!clsu1sun1lsm6 minus letter	
to!clsu1sun1lsm6	L24D	opera
clsu2sun1lst6	L35T	tuam.



to!clM	L77C	corde
to!cl>1	L77C	tuum,
to!po	L84B	Appropinquaverunt
to!pohd	L105A	quaeritis,
to!to *Cf.*	to!toppn1sun1 minus 2x uncinus	
to!tosun1 *Cf.*	to!toppn1sun1 minus uncinus	
to!toppn1sun1	L43L	discam
to!to!pi!cl *Cf.*	to!to!pi!clpp2 minus prepunctis	
to!to!pi!clpp2	L60A	nostrum,
toS!to	L53C	Confitemini
toS!tolst9	L90D	Deus,
toS!pe	L123E	Exaudi,
to!or	L67S	Domine.
to!orsun1lsa5	L8U	Domine.
to!orlst3	L114V	audiatur,
to!orlsc6	L152S	taurorum,
to!pi!cl	L31D	est,
to!pi!cl!pi!pi	L40Q	millia
to!pi!cl!pi!orsun1lsa5	L40Q	speravit,
oc!vi *Cf.*	oc!visun1 minus uncinus	
oc!visun1	L95D	Domine,
orsu1sun1	L94C	Domine.
orsu1sun1lsc2	L94C	Domine.
orsu1sun1lsc3	L94C	iniquis
un!clG	L95D	exaudi
topp2	L95D	meus
topp2sun1	L45R	tuarum,
toppn3ppz1sun1	L58A	me.
to>pp2	L90D	insipientiam
pupp2su1sun1	L95D	sunt.
pupp2su2	L93E	me.
pupp1su2	L64O	opportuno.
pupp1su1sun1	L58A	clamavi,
puppn1su1sun2	L40Q	mille,
puppz1su1sun2	L40Q	mille,
puppz1su2	L168S	tua
ds1!ds1	L43L	meos,
ds1!ts	L67S	mecum
ds1!ts1	L60E	tuo.
cl1!orsun1	L43L	considerabo
cl1!orsun1lsa5	L8U	fac
cl!or *Cf.*	cl!orpp2 minus prepunctis	
cl!orpp2	L93G	induebam
cl1!orpp2	L103R	mirabilis
ds1!clsun2 *Cf.*	ds1!clsun2lst5 minus letter	
ds1!clsun2lst5	L43L	doceas
ds1!clsu1sun1 *Cf.*	ds1!clsu1sun1lst5 minus letter	
ds1!clsu1sun1lst5	L56S	Salvum
ds1pp3	L149E	tibi.
vi>1pp2	L67S	umbrae
vi>1ppz1ppq1	L40Q	custodiant
vi>1ppn1ppz1ppq1	L106S	Alleluia.
pfsun1 *Cf.*	pfsun1lsn5 minus letter	
pfsun1lsn5	L67S	tua,
pfppn1ppq1sun1lsa5	L123A	Deus
un1ppn1ppq1sun2	L112O	estis,
un1sun2 *Cf.*	un1ppn1ppq1sun2 minus uncinus & quilisma	
un1ppn1sun2	L112O	Christum
un1ppn1sun1su2lsa5	L152I	accelera,
clsun1lst5	L44I	vespertinum.
clsu3lst5	L60E	tuo.
tosu1sun1 *Cf.*	tosu1sun1lsn6 minus letter	
tosu1sun1lsc6lst7	L109P	vocavit
tosu1sun1lsn6	L19B	venit
tosu2 *Cf.*	tosu2lst6 minus letter	
tosu2lst6	L37M	tu
tgsun1lsa5	L103H	quoniam
cl!cl!clsun1lsa5	L103P	Alleluia.
cl!cl!cl *Cf.*	cl!cl!cllsn9 minus letter	
cl!cl!cllsn9	L31D	Domine
cl!cl!pi	L54B	commovear
cl!cl!pi!vi	L123A	Dominus
cl!cl!to	L149E	meam,
cl!cl!po	L81R	insurgentibus
cl1!pi!pi!vi	L54B	es
cl1!pi!pisun1lsa5	L174O	plaudite
cl1!pi!tosu1	L37S	mei
cl1!pi!tosun1	L174O	plaudite
sa1sun1lsa5	L11Q	tuam
trsun2 *Cf.*	trsun2lsa5 minus letter	
trsun2lsa5	L13A	Ave
trsux1 *Cf.*	trsux1lsa5 minus letter	
trsux1lst8	L20P	eius,
trsux1lsa5	L136A	aeternam
trsu2 *Cf.*	trsu2lst3lsm8 minus letters	
trsu2lst3lsm8	L136A	vobis
trsun1su2 *Cf.*	trsun1su2lsa3 minus letter	
trsun1su2lsa3	L149C	Domini
trsun1lsa3lsa5	L18L	terra
trsun1lsa5 *Cf.*	trsun1lsa3lsa5 minus letter	
bvsun4	L8U	me.
oc!cl	L31D	Domine.
oc!clM	L77S	peccatorum.
oc!pe!pu	L53C	tuo.
tg!cl	L31D	est,
tg!cl>1	L33C	eius,
tg!cllsn9	L68Q	Sion
tg!or	L77C	tibi
unppn1sun2	L40Q	Cadent
unppn1sun2lsa5	L123E	quaesivi
unppn1sux1	L72B	eius
unppn1sux1lsa5	L24D	exaltavit
unppn1sun1su2	L58A	Domine,
unppn1ppz1sun1 *Cf.*	unppn1ppz1sun1lsa5 minus letter	
unppn1ppz1sun1lsa5	L80S	habitat
unppn3sun1lsa5	L24D	fecit
ta>ppn1	L95D	et
ta>pp2	L33C	ut
pt1ppn1	L99P	tuo.
tssun1lsa5	L24D	dextera
cl>1pp2	L114C	alleluia,
tasu2	L24T	tuam.

significative letters

lsa *Cf.*	vi>lsa6		augete
lsc *Cf.*	pe!unlsc2		celeriter
lseq	L78H	hic	equaliter
lseq-	L78H	quotiescumque	equaliter
lsequ	L56D	confundantur	equaliter
lsf *Cf.*	polsf2lst3		fastigium
lsh	L44D	a	humiliter
lshn *Cf.*	cllshn8		humiliter nectum
lshp *Cf.*	trlst3lshp7		humiliter parum
lsl *Cf.*	pelsl2		levare
lsn *Cf.*	cllsn9		non (tenere), negare, nectum, naturaliter
lsnl *Cf.*	clGlsnl2		non levare
lsnt *Cf.*	pulsnt3		non tenere
lsm	L67S	Virga	mediocriter
lsmd	L82O	da	mediocriter
lss	L95D	tempus	sursum
lssimp	L8U	Domine,	simpliciter
lssimpl	L81R	insurgentibus	simpliciter
lssp	L54Ae	Domine,	sursum parum
lsst *Cf.*	pelsst1lsm4		sursum tenere
lst *Cf.*	clsun1lst5		tenere
lsth *Cf.*	tolsth8		tenere humiliter

tironian notes

lti	L78H	pro	iusum
ltdo	L14O	misericordiam	deorsum
ltdr	L132E	descendit	devertit
ltdx	L122V	intuerentus	devexum
ltps	L128S	veniat,	prode sub eam (trade subtus)
ltqm	L118E	sum	quam mox
ltsb	L11Q	et	sub
ltse	L144C	corde	seorsum
ltsj	L164D	meam	subjice
ltsl	L117R	delectati	saltim
ltsn	L17H	gloriam	sonare
ltsp	L10I	Ierusalem	supra
ltsr	L14P	et	sursum
ltst	L25L	omnes	saltate (salte)
ltus	L17H	eius.	ut supra